

Marital Stability and Intrahousehold Inequality

Tomoki Fujii^{*} Xirong Lin[†] Jacob Penglase[‡]

August 20, 2026

Abstract

We study what predicts marital instability using a unique dataset from Japan with detailed measures of spouses' consumption, savings, labor supply, satisfaction, and marriage-market conditions. We combine descriptive survival and event-study analyses with several econometric and machine-learning approaches, including Cox proportional-hazards models and DynForest, a method designed for survival data with time-varying covariates. Across these approaches, match quality, household resources, time use, stated satisfaction, and marriage-market conditions all contain predictive information. Our main contribution is to show that intrahousehold consumption allocation matters, highlighting information that is missed by standard demographic, time use, and income measures. We relate these findings to the theoretical literature on the drivers of divorce.

JEL Codes: J12, D12, D13, C53.

Keywords: Divorce, Marital Stability, Intra-household Allocation, DynForest.

^{*}Singapore Management University, (e-mail: tfujii@smu.edu.sg).

[†]Emmanuel College, Department of Economics, 400 Fenway, Boston MA, 02115 (e-mail: linx2@emmanuel.edu).

[‡]San Diego State University, Department of Economics, 5500 Campanile Drive, San Diego CA, 92182 (e-mail: jpenglase@sdsu.edu).

We thank Arthur Lewbel, Jianfei Cao, Xiangdan Piao, Miki Kohara, and participants of the AEA Annual Meeting, the Japanese Economic Association Annual Meeting, and the Chinese Economic Society Summer Meeting (SWUFE). All errors are our own.

1 Introduction

Having a stable marriage, where both partners know they are better off remaining married rather than separating, shapes how couples make economic decisions. The perceived risk of divorce may affect labor supply, savings, and resource allocations within the household, as uncertainty may limit the gains from marriage. Given the economic importance of these decisions and rising divorce rates in parts of the world, understanding the determinants of marital instability is increasingly relevant for both researchers and policymakers.

Marital stability, however, is not directly observable. While survey-based closed-ended questions have been used (e.g., [Booth *et al.* 1983](#)), there is no universally accepted measure. In contrast, divorce is an observable outcome that provides a revealed measure of underlying marital instability. In this paper, we identify predictors of divorce—and by extension, marital instability—using a range of methods from semi-parametric ones, such as the Cox proportional hazards model, to non-parametric techniques like random forests. Our key contribution is to identify predictors of divorce using panel data and complementary survival and machine learning approaches, with a novel focus on intra-household consumption allocation.

Studies on the drivers of divorce have tended to be theoretical and existing empirical studies remain fragmented and typically focus on specific channels.¹ We begin by reviewing this literature, starting with seminal work by [Becker *et al.* \(1977\)](#) and [Becker \(1981\)](#), and continuing with [Weiss & Willis \(1997\)](#), among others. We highlight the potential roles of match quality, imperfect information, marriage-market conditions, unexpected shocks that affect the balance of power within the household, and consumption inequality, and use these insights to guide our prediction analysis. These theoretical insights directly inform our empirical specification and interpretation of predictive patterns.

For all of our analyses, we use the Japanese Panel Survey of Consumers (JPSC), a unique dataset designed to study issues related to gender and intra-household inequality, while also including standard labor supply and demographic information. Notably, the survey includes individual savings and consumption measures, allowing us to determine the importance of intra-household dynamics in predicting divorce. For example, are couples with unequal consumption levels more likely to divorce? Are large public goods expenditures associated with more stable marriages?² In addition, the survey includes measures of marital satisfaction, which have previously been shown to be predictive of divorce in Germany by [Arpino *et al.* \(2022\)](#) using similar methods to ours. Finally, the panel nature of the data allows us to make predictions that account for the time-varying nature of individual and household characteristics.

We begin our empirical analysis by descriptively comparing divorced and never-divorced couples using Kaplan-Meier survival curves, and event study methods, where we measure changes in the behavior of couples in the years leading up to divorce. The Kaplan-Meier curves show that couples in which both spouses have high levels of education are the most likely to remain married, while couples in which both spouses have low levels of education are the least stable. The event-study results provide further evidence that couples' behavior changes shortly before divorce as couples reduce spending on public goods and increase the wife's share of consumption, and we observe a pronounced shift in time use for wives towards market work.

¹Past empirical studies have investigated the gap in happiness between spouses ([Guven *et al.*, 2012](#)), economic gains and the non-pecuniary quality of the match ([Chiappori *et al.*, 2018](#)), personality traits ([Lundberg, 2012](#)), negative shocks to men's earnings ([Weiss & Willis, 1997](#)), job displacement and disability ([Charles & Stephens, 2004](#)), and legal and informational constraints ([Peters, 1986](#)). See [Amato \(2010\)](#) for a comprehensive literature review on the empirical determinants of divorce.

²[Lise & Yamada \(2019\)](#) similarly study the behavior of married couples using the JPSC and a dynamic collective household model. Our study complements their findings by taking a more descriptive approach to identifying key factors that impact marital decisions. [Kureishi *et al.* \(2025\)](#) also use the JPSC to study the impact of unexpected labor market shocks—specifically the 2011 earthquake in Japan—on divorce, as well as labor and consumption allocations within the household.

We then turn to several econometric and machine learning techniques to examine the predictors of divorce. Given the presence of right-censoring in marriage duration data, we employ methods tailored to survival analysis. Our first approach is Lasso Cox, which combines the interpretability of the standard Cox proportional hazard model with flexible variable selection. This approach is well suited to our setting because it allows us to identify the most important predictors of divorce while retaining the structure and interpretability of a survival model.

We also estimate DynForest, a machine learning algorithm developed by [Devaux *et al.* \(2023\)](#) for analyzing time-to-event data. Like random survival forests ([Ishwaran *et al.*, 2008](#)), DynForest is an ensemble method based on decision trees, but it can also accommodate time-varying predictors from panel data. As discussed in [Wang *et al.* \(2019\)](#), machine learning methods often achieve higher predictive accuracy than parametric or semi-parametric alternatives, particularly with high-dimensional data or violations of the Cox proportional hazards assumption. However, non-machine-learning approaches offer greater interpretability and can accommodate covariates that DynForest cannot, such as time-varying discrete variables. Since our goal is to identify predictors of divorce rather than compare model performance, we present results from all approaches.

In our Lasso Cox specification, we consider 73 characteristics of the couple spanning a wide range of dimensions, including consumption allocation, time-use decisions, demographics, marriage market conditions, shocks, and stated satisfaction with different aspects of life. The Lasso Cox procedure selects predictors from across all these dimensions, indicating that the factors associated with divorce hazard are broad-based rather than concentrated in any single domain. Consistent with [Arpino *et al.* \(2022\)](#), the wife’s satisfaction with the relationship is negatively associated with divorce hazard, suggesting that lower relationship satisfaction is predictive of marital instability. The model also selects measures of household resources and allocations, including the wife’s income, the wife’s market work, home ownership, the public-goods budget share, and various characteristics of the couple’s parents. In addition, important events in the wife’s life, particularly negative events, are associated with higher divorce hazard. These patterns suggest that both subjective assessments of the marriage and time-varying economic behavior are informative predictors of divorce.

The DynForest model allows us to flexibly incorporate nonlinear relationships between the outcome and predictors and include predictors that change gradually overtime. The variable-importance results show that several of the most predictive individual variables are time-varying characteristics, including the wife’s income and leisure hours, the couple’s number of children, and several measures of income. Satisfaction and marriage-market variables are also predictive, especially the wife’s satisfaction with her relationship. Taken together, the Lasso Cox and DynForest results suggest that divorce is predicted by a combination of relationship quality, household resources, time use, intra-household allocation, marriage-market conditions, and some characteristics measured at the time of marriage, rather than by any single factor alone.

Our paper makes three main contributions. First, we incorporate detailed intra-household consumption allocation data into the analysis of marital stability. While demographic and labor-supply variables are commonly observed in household surveys, information on how expenditures are allocated across spouses, children, and other household public goods is rarely available. Using the JPSC, we examine the importance of consumption allocations across spouses as well as the share of consumption devoted to public goods (including children). In our preferred results using a Lasso Cox specification, we find the share of consumption devoted to public goods to be a significant predictor. These results are confirmed in our DynForest specifications using group variable importance (gVIMP), which accounts for collinearity among predictors.³

³Previous work by [Piao \(2021\)](#) documents the relationship between the wife’s income share, private consumption, and marital stability in

Second, we contribute to the empirical literature on divorce prediction by combining survival models with machine-learning methods suited to right-censored marriage-duration data. In particular, we estimate Lasso Cox models and DynForest, a random-survival-forest method developed by [Devaux *et al.* \(2023\)](#) that incorporates longitudinal covariates. This extends the approach of [Arpino *et al.* \(2022\)](#), who use random survival forests to study union dissolution, by allowing us to exploit the panel structure of the data and examine how time-varying predictors—such as income, labor supply, satisfaction, and consumption allocations—are associated with divorce hazard. The two approaches offer complementary advantages. Lasso Cox provides an interpretable framework for selecting predictors of divorce hazard from a large set of covariates, including discrete events such as shocks. DynForest, in turn, can exploit the trajectories of repeatedly measured covariates (such as consumption, savings, and labor supply) while flexibly accommodating nonlinearities and interactions in their relationship with divorce risk. By comparing the predictors identified by these complementary approaches, we assess which factors emerge consistently as important predictors of divorce.

Third, we study divorce in Japan, an important and relatively understudied setting in the empirical divorce literature. Japan provides a distinctive institutional and cultural context in which divorce norms, marriage rates, women’s labor-market behavior, and family arrangements have changed substantially over time. We incorporate marriage-market characteristics into the prediction exercise and examine how these factors relate to divorce risk. In doing so, our study complements recent work on marriage, divorce, and intra-household allocation in Japan, including [Piao \(2021\)](#) and [Kureishi *et al.* \(2025\)](#).

It is important to note that our approach is descriptive rather than causal. While a number of existing empirical studies estimate the causal effects of various shocks (e.g., job loss or health events) on divorce or structurally model them, our goal is to use predictive methods to uncover underexplored correlates of marital instability. This allows us to better understand which factors, including those not commonly observed or studied, are most associated with divorce. Our findings help inform future theoretical work on marital instability and support the view of [Lewbel \(2019\)](#), who argues that the correlations uncovered through machine learning exercises heighten the need for structural economic models to explain them.

The remainder of the paper is organized as follows. In [Section 2](#) we discuss the theoretical literature on marital stability and empirical studies that have tested these theories. [Section 3](#) describes marriage and divorce in Japan. We then discuss the data in [Section 4](#), and conduct several descriptive analyses before proceeding to our more formal prediction exercises in [Section 5](#). [Section 6](#) concludes.

2 Theoretical Framework

In this section, we briefly review the main theoretical and empirical literature on marital stability, which informs the selection of couple characteristics potentially relevant to divorce. We then present a parsimonious analytical framework that motivates the empirical predictions. We begin with the standard view of marriage in which divorce occurs when the value of remaining married falls below the value of separation ([Becker, 1974](#); [Becker *et al.*, 1977](#)). This comparison depends on match quality, household resources and allocations, shocks to households, and marriage market conditions, among others. In the next subsection, we review these elements. We then bring these elements together in a unified analytical framework Japan. Our empirical approach, however, is significantly different.

and discuss the empirical predictions in the subsequent subsection.⁴

2.1 Factors Associated with Marital Stability

Becker (1974) and Becker *et al.* (1977) emphasize that marriage is sustained by gains that are specific to the relationship. These gains may arise from various forms of marriage-specific capital. One important form is physical capital, including housing and other jointly held assets. The accumulation of such capital can strengthen marital stability by increasing the value of remaining together and by facilitating specialization within the household (Rainer & Smith, 2010; Farnham *et al.*, 2011; Lafortune & Low, 2017, 2023). Second, children are another important form of marriage-specific capital. They may play a role similar to physical capital, because the value of child-related public goods is typically higher when the parents remain together than after separation (Chiappori *et al.*, 2015). A large empirical literature finds that children, especially young children, stabilize marriage (Waite & Lillard, 1991; Kalmijn & Poortman, 2006), though such effects may depend on the children’s gender (Dahl & Moretti, 2008; Kabátek & Ribar, 2021).⁵ Third, match quality is also an important component of marriage-specific capital. Couples who are similar in their characteristics—such as age, education, income, race, religion, and family background—may have more stable marriages due to higher expected marital surplus (Becker *et al.*, 1977). Existing empirical studies are broadly consistent with this idea (Weiss & Willis, 1997; Lehrer & Chiswick, 1993; Lee & McKinnish, 2018; Zhang & Van Hook, 2009).

Besides marriage-specific capital, the attractiveness of staying married may change due to unforeseen shocks (Becker *et al.*, 1977), as they alter the trajectory of their expected incomes, savings, roles in household production, intrahousehold resource allocations, fertility decisions, and outside options. In dynamic models of couple behavior, such as the collective household model, decisions are typically characterized as the outcome of bargaining between spouses, with intrahousehold resource allocations determined by individual participation constraints and relative bargaining power (Manser & Brown, 1980; McElroy & Horney, 1981; Chiappori, 1988, 1992; Lundberg & Pollak, 1993). By affecting spouses’ outside options or other determinants of bargaining power, unforeseen shocks can shift these bargaining weights and thereby change the allocation of resources within the household. Such changes may, in turn, affect the gains from remaining married and ultimately the risk of divorce.

Empirical studies are broadly consistent with this idea. Negative economic shocks such as job losses have been found to increase divorce risk (Weiss & Willis, 1997; Velilla *et al.*, 2025; Charles & Stephens, 2004; Marinescu, 2016). Changes in consumption and time in household production may also be predictive of subsequent divorce. Olafsson & Steingrimsdottir (2020) find that a government program that expanded paternity leave and husbands’ involvement in childcare reduced divorce among eligible couples. This finding is consistent with Kalmijn (1999), who finds that the wife’s marital satisfaction is higher when her husband is more involved in childcare.

Given our data, the allocation of consumption within the household is especially relevant to our setting. Models based on revealed preference and collective household behavior show that consumption allocation and marriage-market conditions are informative about marital stability (Cherchye *et al.*, 2021, 2025). The JPSC is unusual in measuring expenditures separately for the wife, husband, children, other household members, and the household as a whole. This allows us to

⁴For survey articles on the causes of divorce, see Lyngstad & Jalovaara (2010). Recent work in this area includes Rosenfeld (2018), Abdel-Sater (2022), and Rosenfeld & Roesler (2024), among others.

⁵See also other related studies such as Cherlin (1977), Heaton (1990), and Svarer & Verner (2008).

examine whether private consumption shares and public-good expenditure shares predict subsequent divorce, rather than relying only on aggregate household expenditure.

Changes in the marriage market also alter the prospect of remarriage and thus the value of outside options. Marriage-market conditions are therefore key determinants of divorce decisions. Consistent with this idea, [South & Lloyd \(1995\)](#) find that the larger the number of potential partners in an individual’s marriage market, the higher the likelihood of divorce. [Chiappori *et al.* \(2002\)](#) show theoretically and empirically that the sex ratio in each spouse’s marriage market can shift bargaining power within the marriage and affect divorce decisions. Finally, subjective measures such as happiness and relationship satisfaction may capture otherwise unobserved shocks to marriage. These are important as psychological well-being, personal traits, and spousal differences in happiness are found to be associated with marital outcomes ([Gardner & Oswald, 2006](#); [Lundberg, 2012](#); [Guven *et al.*, 2012](#)).

Based on the discussion above and data availability, we divide our predictors in the empirical analysis into five broad categories: match quality, household resources and allocations, shocks, marriage-market conditions, and satisfaction. Match-quality variables capture marriage-specific capital, while the remaining categories primarily reflect changes in bargaining positions, changes in outside options, and shocks to the marriage, albeit imperfectly. These mechanisms have often been studied in separate strands of the literature. We therefore introduce below a simple unified framework that links them and clarifies the empirical predictions that guide our analysis.

2.2 A Unified Analytical Framework and Predictions

This subsection introduces a unified analytical framework that combines the key elements emphasized in the divorce, household-bargaining, and marriage-market literatures reviewed above. These strands of literature respectively emphasize the comparison between the value of marriage and separation, the division of resources within marriage, and the role of outside options and threat points. We also incorporate marriage-specific capital, which conceptually includes children, housing, and shared investments. We bring these elements together in a dynamic bargaining structure, where the current choices within the household affect future marital value. This structure is broadly consistent with earlier works ([Lundberg & Pollak, 2003](#); [Mazzocco, 2007](#); [Mazzocco *et al.*, 2014](#); [Voena, 2015](#)).

The proposed framework helps organize the empirical exercise and clarify why the observed variables—marriage-market conditions, satisfaction, labor supply, savings, consumption allocation, and public-good expenditure—should be informative about divorce hazards. In the main text, we discuss the main building blocks of the dynamic model of bargaining and divorce we consider and the empirical predictions derived from the model. The mathematical details of the model are relegated to [Appendix C](#).

We consider a married couple—consisting of husband h and wife w observed in discrete time. At the beginning of each period t , the couple is characterized by four state variables. First, K_t denotes marriage-specific capital, which summarizes shared investments that are more valuable inside the current marriage than outside it, such as children, housing, and other joint investments. Second, ε_t denotes match quality, which is not directly observable but may be reflected in reported relationship and other satisfaction measures. Third, $Y_t = (Y_{ht}, Y_{wt})$ denotes a vector of spouses’ incomes. While we refer to Y_t as income, it could conceptually include other time-varying economic resources. Fourth, $M_t = (M_{ht}, M_{wt})$ denotes spouse-specific marriage-market conditions, where a higher value represents better prospects after divorce.

Conditional on the state vector $S_t \equiv (K_t, \varepsilon_t, Y_t, M_t)$, spouses determine the intrahousehold allocation of resources (C_{ht}, C_{wt}, G_t) subject to the budget constraint through Nash bargaining, where C_{it} is private-good expenditure for spouse $i \in \{h, w\}$ and G_t is public-good expenditure. Each spouse also has an outside option V_{it}^O that depends on own resources and marriage-market conditions. The marriage continues only if there exists a feasible allocation that gives both spouses utility greater than their outside options. Divorce occurs when no feasible allocation can satisfy both spouses' participation constraints. Based on the framework outlined here, we derive the following four predictions:

Prediction 1 (Outside options and divorce risk). *An improvement in a spouse's marriage-market conditions weakly increases divorce risk.*

This prediction follows from the participation-constraint logic in Becker (1974) and Becker *et al.* (1977). Holding the value of marriage fixed, better prospects outside marriage raise the spouse's threat point and make it harder to find an allocation that keeps both spouses in the marriage. It is also consistent with marriage-market models in which sex ratios and remarriage prospects affect bargaining and marital stability.

Prediction 2 (Match quality and marital stability). *Higher match quality raises the value of marriage. Therefore, adverse shocks to match quality reduce marital value and increase divorce risk. Since (most) shocks to match quality are not directly observed, lower relationship satisfaction should be associated with a higher divorce hazard.*

This prediction restates the standard idea that divorce becomes more likely when realized match quality is worse than expected. In the empirical analysis, relationship satisfaction and related subjective measures serve as noisy proxies for match quality, while negative life events may capture shocks that lower current or expected future marital value.

Prediction 3 (Shared investment and divorce hazard). *Lower public-good expenditure today reduces future marriage-specific capital and lowers future marital value. Therefore, lower expenditure on public goods should precede divorce and be associated with a higher divorce hazard.*

Expenditure on children and other household public goods is not only part of current consumption; it can also sustain future marriage-specific capital. A decline in such expenditure may therefore signal weaker investment in the joint future of the marriage and should be predictive of divorce. This prediction is especially relevant for our setting because the JPSC separately measures expenditures for the wife, the husband, children, other household members, and the household as a whole.

Prediction 4 (Outside options matter more when match quality is low). *The effect of outside options on divorce risk is stronger when match quality is low. Therefore, marriage-market conditions should have a larger effect on divorce hazard among couples with lower reported satisfaction.*

The intuition is that better marriage-market conditions raise outside options in all states, but this increase is most likely to trigger divorce when the value of staying married is already low. In empirical terms, if higher satisfaction indicates higher match quality, favorable marriage-market conditions should be more strongly associated with divorce among less satisfied wives. Equivalently, if higher values of the marriage-market variable indicate better outside options and higher satisfaction indicates higher match quality, the positive association between marriage-market conditions and divorce hazard should be weaker at higher levels of satisfaction.

The four predictions guide the empirical analysis in Section 5. Predictions 1 and 2 organize variables that are already prominent in the divorce and bargaining literatures. Predictions 3 and 4 offer additional implications of combining dynamic marriage-specific capital with intrahousehold bargaining and outside options. They motivate us to include variables on consumption allocation, public-good expenditure, and interactions between satisfaction and marriage-market conditions in the set of potential predictors of divorce.

3 Background: Marriage, Divorce, and Family in Japan

Empirical studies on marriage and divorce in Japan are relatively scarce compared to those for the US or Europe. Japan offers a distinct institutional and cultural setting, as its family system is neither purely traditional nor purely modern, but instead combines elements of both (Kumagai, 2015). In particular, contemporary Japanese families reflect features of both nuclear families consisting of parents and children, and modified stem families, in which multiple generations may co-reside under a hierarchical structure.

Over the past decades, patterns of formation and dissolution in Japan have changed substantially. Nearly 70 percent of marriages were arranged in the 1930s, but this share declined rapidly over time. The proportions of arranged and romantic marriages equalized in the late 1960s, and the latter accounted for the vast majority of marriages by 2010. Notably, the meaning of “arranged” has also evolved, shifting from parental arrangement toward facilitated introductions (Tokuihiro, 2009).

Another important trend is that marriages are occurring later and less frequently. Between 1975 and 2020, the mean ages at first marriage increased from 27.0 to 31.0 for men and 24.7 to 29.4 for women, respectively. Over the same period, the share of individuals never married by age 50 rose from 2.1 percent to 28.3 percent for men and from 4.3 percent to 17.8 percent for women. Retherford *et al.* (2001) argue that the erosion of arranged marriage contributed to these changes.

Japan’s divorce patterns have also evolved. The crude divorce rate (CDR), defined as the number of divorces per 1,000 people in a year, was historically high, reaching nearly three in the late 19th century, likely reflecting the prevalence of arranged marriages and the relative ease of divorce at the time, among other reasons (Kumagai, 1983; Nishida *et al.*, 1987). Following the introduction of the modern Civil Code and Household Registration Act in 1898, the CDR declined and reached 0.68 by 1940. After the Second World War, it gradually increased, peaking at 2.3 in 2002, before declining to 1.5 in 2021. However, the CDR is not a direct measure of marital stability, as lower marriage rates mechanically reduce the CDR even if the conditional probability of divorce remains unchanged. Cohort-based evidence suggests that the proportion of ever-married individuals who experience divorce has increased over time (Iwasawa, 2008).

In contemporary Japan, most divorces occur by mutual agreement rather than through judicial procedures. As of 2021, 86.4 percent of divorces were by mutual agreement, although this share has declined somewhat over time. The distribution of marriage duration at divorce has also changed. Early divorces within four years have declined as a share of total divorces, while divorces among longer-duration marriages have become more common. This pattern partly reflects demographic changes due to the aging of baby boomers, but may also be associated with declining social stigma and policy changes, such as pension-splitting rules.

A growing literature examines factors associated with marriage and divorce in Japan. Kaufman & Taniguchi (2010) find that married women report higher levels of happiness, while Lim & Raymo (2016) document links between marriage, labor

supply, and health outcomes. Social norms have also evolved: attitudes toward women’s work have shifted over time and across cohorts (Lee *et al.*, 2010; Yoshida, 2010).

Education plays an important role in marriage patterns. Using data from the Japanese National Fertility Surveys, Fukuda *et al.* (2020) document a reversal in the education gradient in women’s age at first marriage—from a negative to a positive gradient—driven by declining marriage rates among less-educated women and increasing rates among more-educated women.

Finally, Japan’s divorce rate remains relatively low compared to other high-income countries. While cross-country comparisons are imperfect, several factors may contribute to this pattern. Katsurada (1997) argues that the costs of divorce may be higher for Japanese women due to greater childcare responsibilities and weaker labor market prospects after divorce. Social norms surrounding family roles may further reduce the perceived benefits of divorce.

These factors may help explain Japan’s relatively low observed divorce rates and provide important context for interpreting our empirical results, particularly regarding the roles of consumption allocation, labor supply, and marriage market conditions in shaping divorce behavior.

4 Descriptive Analysis

4.1 Data

We use the JPSC data from 1993 to 2020, which include information on women and their families, with an emphasis on household economics, employment, and relationships.⁶ The first wave of the JPSC was conducted in 1993 with 1,500 women aged 24–34 years. Since then, it has been conducted every year, with younger cohorts of women added approximately every five years. This allows us to observe the same women at different life stages (e.g., 20s, 30s, 40s) over time. As of 2018, respondents in the 26th round were aged 29 to 59.

Our unit of observation is a couple-year and our sample consists of married-couple households, with and without children. We retain households that co-reside with parents or parents-in-law, but exclude observations where respondents co-reside with siblings, other relatives, or non-relatives. After dropping couple-years with missing data for any of the covariates (discussed below), the final sample consists of 2,189 couples with 19,333 couple-year observations (18,208 for never-divorced couples and 1,125 for divorced couples).⁷ In our sample, 166 couples divorced during the observation period. The average marital duration for these couples is about 15 years, and three-quarters of all divorces occur before the 22nd year of marriage.

Throughout the prediction analysis, we impose a one-period timing convention between the covariates and the divorce outcome. For each couple-year t , we use information observed up to and including period t to predict whether the couple divorces in period $t+1$. Thus, the predictors are measured while the couple is still married and precede the divorce outcome they are used to predict.

Below, we discuss several key categories of variables that comprise our predictors. These include standard demographic data from the JPSC, as well as several novel aspects of it, such as individual consumption and savings. We also discuss how we construct key marriage market variables. Additional details are provided in the Appendix.

⁶Available from the Panel Data Research Center at Keio University. Detailed information on the JPSC data is available on its website (<http://pdr.c.keio.ac.jp/en>).

⁷Two women married twice in this sample. For these women, we treat each marriage separately.

Demographic Information. The JPSC contains rich information on the demographic characteristics of the respondents and their parents.⁸ We interpret these variables as reflecting the match quality of the marriage. There is also non-standard information regarding how many female and male friends the wife has, and we consider these questions as reflective of the strength of the wife’s social network.

Consumption, Savings, and Labor Supply. The JPSC features individual-level consumption spending and savings data. The expenditure data are particularly novel as they include both standard household-level spending on food, clothing, education, healthcare, and individual-level data measuring person-level consumption. Expenditures are classified as either (i) expenditures for the wife, (ii) expenditures for the husband, (iii) expenditures for children, (iv) expenditures for other household members, and (v) expenditures for the household as a whole. We interpret (i)–(ii) as private goods expenditure and (iii)–(v) as public goods expenditure. As highlighted in [Becker \(1974\)](#) and [Stevenson & Wolfers \(2007\)](#), consumption complementarities are an important driver of marriage. Survey questions on individual savings are structured similarly and allow us to see if spouses who anticipate future divorce increase their savings as an insurance mechanism. In addition, the individual-level time use data are also included in the JPSC, which we categorize as either household work, work in the market, or leisure time.

Income Control. In addition to private consumption and savings, the JPSC also asks a series of questions about the management of household income. From these questions, we are able to determine the proportion of income earned by each spouse that is controlled by the couple either individually or jointly. See Appendix Section [A.1](#) for details.

Women’s Satisfaction Measures. For women, we observe their self-reported happiness level and satisfaction with various aspects of life, including relationships, life, income, and household spending. These measures provide us with an additional measure of marital stability that we employ in our empirical analyses. Respondents choose levels ranging from 1 - 5 where 1 indicates the lowest level of satisfaction.

Shocks and Events. The JPSC asks respondents whether they have experienced any of the following events in the past year. The events include: (1) Got a job; (2) Changed jobs; (3) Retired; (4) Entered school; (5) Started a new hobby; (6) Experienced a serious illness; (7) Experienced psychological illness; (8) Experienced consumer troubles; and (9) Experienced an accident or disaster. We group (1) to (5) into “positive or ambiguous events for the wife” and (6) to (9) into “negative event for the wife”.

Marriage Market Data. Marriage market characteristics, such as the availability of suitable new partners in a similar age range should the couple separate, play a crucial role in shaping marital stability. To incorporate this idea into our analysis, we merge marriage market data into our base sample. We follow [Cherchye *et al.* \(2017\)](#) and assume that a man’s potential marriage market includes all marriageable women who are at most one year older and 10 years younger in the survey year, and who live in the same prefecture where he resided at age 18 to construct a proxy for the man’s local marriage market size. For women, we define their potential marriage market similarly, except that potential matches are men between one year younger and 10 years older. These bounds correspond to the fifth and 95th percentiles of the age difference distribution in our sample of couples.

To obtain information on the potential marriage market relevant to each individual, we compile annual data from the

⁸Parental characteristics include education and income. For parental education, we use categorical variables rather than years of schooling for two reasons. First, some of the parents were educated under an older education system prior to the Second World War, and the years of education before and after the old and new education systems are not directly comparable. Second, we have a high prevalence of missingness in parental education variables and therefore decided to have a separate category for this.

Vital Statistics of Japan.⁹ We compiled data on the age at first marriage from 1980 to 2020, as well as marriage and divorce rates per 1,000 individuals at the prefecture level, from the e-Stat website.¹⁰ For the size of the potential marriage market, we use prefecture-level estimates of population pyramids available from the Regional Economy Society Analyzing System (RESAS).¹¹ We use the estimated population size for males and females in each five-year age group every five years between 1980 and 2020. The resulting marriage market sizes are presented in Figure A1 in the Appendix, where we plot market size by age group and gender. We observe small differences in market size between men and women on average, except for individuals aged 20 to 29. In addition, we find significant variations across couples, as illustrated by the large interquartile range across the age distribution.

Further details about the questionnaire, specific questions and answers, the construction of all the covariates that we use in the estimation, and any imputation that we did for the covariates are described in Section A.1 in the Appendix.

4.1.1 Panel Entry and Censoring

Because the JPSC follows both single and married respondents, couples enter our analysis at different points in their marital histories. Some marriages begin after the respondent enters the panel, while others are already ongoing at the respondent’s first observation. For marriages that are already ongoing at entry, we observe the year of the current marriage but not the time-varying covariates for the years prior to survey entry. These marriages therefore contribute left-truncated spells. In our sample, 368 couples marry after entering the survey, while 1,821 are already married at entry. As a result, marriages that both began and dissolved before survey entry are not observed.

This structure raises a standard delayed-entry problem in survival analysis. Cox models with delayed entry (which we use later) can handle such spells by including each marriage in the risk set only after it becomes observable and only conditional on having survived to that duration. This approach yields valid post-entry hazard-ratio estimates under a conditional independent-truncation assumption: among marriages with the same duration and observed covariates, the timing of survey entry is unrelated to subsequent divorce risk. If this assumption fails—for example, because unobserved marital instability affects whether a marriage survives long enough to be observed at survey entry—then the observed left-truncated spells may represent a positively selected group of marital survivors, and estimated divorce hazards may be biased.

Table 1 compares characteristics at the time of marriage for couples already married when the respondent first enters the survey with marriages that begin after panel entry. The table shows that the two groups differ in several ways. Couples in the left-truncated sample marry at younger ages: wives marry at age 24.8 on average, compared with 30.7 in the non-truncated sample, while husbands marry at age 27.3 compared with 32.0. Left-truncated couples also come from earlier birth cohorts and have slightly lower educational attainment at marriage. These differences largely reflect the structure of the JPSC sampling design, in which many respondents were already married at first interview. They also imply that the left-truncated sample contains marriages that survived long enough to be observed and may therefore underrepresent marriages with high early divorce risk.

⁹<https://www.mhlw.go.jp/english/database/db-hw/outline/index.html>

¹⁰<https://www.e-stat.go.jp/stat-search/files?page=1&toukei=00450011>

¹¹The population data in RESAS is only available for five-year age groups, e.g., 20–24, 25–29, etc. We assume a uniform distribution within each age group and divide the group-level population by five to impute the population for each individual age within the group.

Table 1: Left-truncated vs. Non-truncated Samples

	Left-truncated (1)	Non-truncated (2)
Wife's Age at Marriage	24.75	30.68
Husband's Age at Marriage	27.34	31.99
Husband Older	0.70	0.56
Wife's Education at Marriage	13.22	14.28
Husband's Education at Marriage	13.49	14.32
Ratio of Spouse's Education	1.00	1.01
Wife's Birth Year	1970.46	1977.39
Husband's Birth Year	1967.88	1976.06
Number of Couples	1,821	368

Note: Left-truncated sample consists of marriages that began before entry into the survey. Non-truncated sample consists of marriages that took place after entry into the survey.

This selection matters for interpreting divorce hazards and cohort comparisons. Because marriages that began and ended before survey entry are absent from the data, observed left-truncated marriages may be more stable than the full population of marriages from the same cohorts. In addition, exposure differs across cohorts. Older cohorts contribute more observations than younger cohorts, for both wives and husbands. Among wives, 46%, 33%, and 21% are born in the 1960s, 1970s, and 1980s, respectively. Respondents who enter the survey at younger ages can also be followed over a longer portion of their marital histories. As a result, older cohorts tend to contribute longer observed marriage spells, while younger cohorts are more likely to be right-censored after shorter marital durations.

These patterns motivate two empirical choices. First, our survival specifications account for delayed entry whenever possible by placing couples at risk only after they are observed in the panel. Second, in a sensitivity analysis in Section 5.1.4, we address left truncation and differential cohort exposure using inverse-probability weights constructed from the predicted probability that a marriage is observed from its start, based on covariates related to survey entry and marital duration (Seaman & White, 2013). These weights are intended to reduce the influence of systematic differences between left-truncated and non-truncated marriages, while recognizing that the JPSC does not contain complete marriage histories.

4.2 Summary Statistics

We present summary statistics in Table 2. This table includes the variables used in the prediction exercises, such as measures of match quality, household resources and allocations, wife's satisfaction measures, shocks, and marriage market characteristics.¹² Across each panel, we compare characteristics of couples who did not divorce during our observation period (*never-divorced couples*) with those that got divorced (*divorced couples*).

Panel (A) of Table 2 reports measures of match quality, or the characteristics of the couple at the time of marriage. Most differences between never-divorced and divorced couples are minor. However, couples that remained married during the observation period tend to have more equal education levels and own a house at the time of marriage. Further, husbands are, on average, older than their wives with an average age difference of about two years.

¹²One caveat is that we omit the shock variables from our DynForest estimation, as it is unable to incorporate time-varying indicator variables.

Table 2: Summary Statistics

	Never-Divorced Couples (1)	Divorced Couples (2)
<i>Panel (A): Match Quality</i>		
Wife's Education at Marriage	13.44	12.93
Husband's Education at Marriage	13.73	12.47
Ratio of Spouse's Education	1.00	1.06
Condo Owner at Marriage	0.11	0.09
Homeowner at Marriage	0.38	0.27
Husband's Age at Marriage	27.97	26.30
Wife's Age at Marriage	25.65	24.55
Husband Older	0.68	0.67
Parental Income (Imputed Missing Values)	4.21	4.41
Father's Schooling Low	0.72	0.76
Father's Schooling High	0.21	0.20
Father's Schooling Missing	0.07	0.04
Father-in-Law Schooling Low	0.35	0.42
Father-in-Law Schooling High	0.08	0.09
Father-in-Law Schooling Missing	0.57	0.49
Mother's Schooling Low	0.77	0.81
Mother's Schooling High	0.16	0.15
Mother's Schooling Missing	0.07	0.04
Mother-in-Law Schooling Low	0.38	0.46
Mother-in-Law Schooling High	0.04	0.04
Mother-in-Law Schooling Missing	0.58	0.50
<i>Panel (B): Household Resources and Allocations</i>		
Log Expenditure (1,000 yen/week)	3.98	3.97
Log Wife's Income	0.98	2.09
Log Husband's Income	5.30	4.33
Income Ratio (Husband/(Husband+Wife))	0.46	0.56
Wife's Weekly Hours in Housework	46.48	44.67
Wife's Weekly Hours in Market Work	26.28	31.56
Wife's Weekly Hours in Leisure and Sleep	95.19	91.67
Husband's Weekly Hours in Housework	7.39	6.88
Husband's Weekly Hours in Market Work	63.88	63.56
Husband's Weekly Hours in Leisure and Sleep	96.69	97.51
Wife's Days Off per week	1.58	1.65
Husband's Days Off per week	1.48	1.42
Difference in Days Off (Husband-wife)	-0.10	-0.23
Children Under 6	0.38	0.33
Children 6 or Over	1.39	1.49
Wife's Share of Total Consumption	0.06	0.07
Husband's Share of Total Consumption	0.14	0.14
Public Goods Budget Share	0.80	0.79
Consumption Ratio (Husband/(Husband+Wife))	0.56	0.55
Share of Income Controlled by both Spouses	0.75	0.72
Share of Income Controlled by Wife	0.08	0.12
Log Wife's Savings (1,000 yen/week)	-1.87	-2.09
Log Husband's Savings (1,000 yen/week)	-1.53	-2.02
Log Family Savings (1,000 yen/week)	-0.08	-0.91

Continued on next page

Table 2 – continued from previous page

	Never-Divorced Couples	Divorced Couples
	(1)	(2)
Savings Ratio (Husband/(Husband+Wife))	0.53	0.54
Number of Female Friends	15.41	15.50
Number of Male Friends	2.51	3.64
<i>Panel (C): Wife's Satisfaction Measures (on a five-point Likert scale)</i>		
Wife's Satisfaction with Relationship	3.46	2.94
Wife's Satisfaction with Life	3.52	3.17
Wife's Satisfaction with Income	3.32	3.07
Wife's Satisfaction with Spending	3.30	3.23
Wife's Happiness	3.91	3.54
<i>Panel (D): Shocks</i>		
Wife Got a Job	0.07	0.10
Wife Changed Jobs	0.04	0.08
Wife Retired	0.05	0.06
Wife Entered School	0.00	0.00
Wife Started a New Hobby	0.04	0.04
Wife Experienced a Serious Illness	0.01	0.03
Wife Experienced Psychological Illness	0.01	0.03
Wife Experienced Consumer Troubles	0.00	0.00
Wife Experienced an Accident or Disaster	0.02	0.03
Positive or Ambiguous Event for Wife	0.29	0.32
Negative Event for Wife	0.05	0.08
<i>Panel (E): Marriage Market Characteristics</i>		
Husband's Marriage Market Size	800,529.94	788,153.31
Wife's Marriage Market Size	946,626.50	935,793.94
Marriage Rate (standardized)	5.66	5.82
Divorce Rate (standardized)	1.97	2.05
Number of Couple-Year Observations	18,202	1,125
Number of Couples	2,023	166

Note: Mean values across couple-year observations. Low parental education is a dummy variable that indicates having 9 or 12 years of education, while high parental education is a dummy variable indicating having 14 or 16 years of education. Income variables include imputed values. Hours, household expenditure, and savings are all per week. The JPSC delineates between savings for (1) all of my family (2) for me (3) for my husband (4) for my children (5) for the others. Savings for family are the sum of (1), (4), and (5). "Share public goods" is the share of expenditure for family, children, and other out of total household expenditure. Homeowner and condo owner are both dummy variables. Market work hours include work, commute, and study hours. Housework hours include both household chores and childcare time. Days off measures number of days off per week. Log Husband (Wife)'s Income is Husband (Wife)'s log annual income in ten thousand yen. Savings and expenditures are in thousand yen. Happiness ranges from value 1–5: 1 "very unhappy", 2 "a little unhappy", 3 "neither happy nor unhappy", 4 "rather happy", 5 "very happy". Satisfaction measures range from value 1–5: 1 "not at all", 2 "a little", 3 "moderately", 4 "quite much", 5 "very much". Event variables are all dummy indicator variables. Shock variables are those that happened in the past year. In Panel (D), we group (1) Got a job (2) Changed jobs (3) Retired (4) Entered School (5) Started a new hobby into "positive or ambiguous event for wife". We group wife experiencing serious illness, psychological illness, consumer troubles and accident or disaster into "negative event for wife". Marriage and divorce rates are measured as marriages and divorces per one thousand people.

In Panel (B), we examine the behaviors of married couples and how they allocate their time and consumption. We find clear evidence of specialization among married Japanese couples. First, the average labor income of men is almost double that of women, despite men and women having similar years of schooling. Second, men spend more hours in market work

and fewer hours on household work—including domestic chores and childcare—than women. However, total working hours (market plus household work) are similar between men and women. Third, work experience at the time of marriage is comparable between men and women. Throughout marriage, husbands accumulate more work experience than wives, as wives allocate more time to part-time jobs and household work after getting married.

A key focus of our analysis is how consumption is allocated and the role that this allocation plays in marital stability. We find a large gender asymmetry in the share of consumption between the husband and wife, with this asymmetry being larger among never-divorced couples. Moreover, we observe greater expenditures on public goods for never-divorced couples. Across both types of couples, we find that husbands have more leisure and sleep hours than their wives. The gender gap in market work is more pronounced in couples who stay married.

A unique aspect of the data is the self-reported satisfaction measures for the women in the sample. Panel (C) of Table 2 reports average satisfaction levels in five different dimensions on a five-point Likert scale, with higher values indicating greater satisfaction or happiness. Unsurprisingly, there is a clear difference in reported satisfaction between women who divorce and those who do not. This difference is especially large for satisfaction with their relationship with their husband (3.46 among never-divorced women and 2.94 among divorced women).

Panel (D) highlights the differences in shocks experienced by wives between never-divorced and divorced couples. Wives in divorced couples are more likely to report employment transitions, such as obtaining a new job (7% for never-divorced vs. 10% for divorced) or changing jobs (4% vs. 8%), as well as negative health events, including serious and psychological illnesses. Although positive or ambiguous life events are similarly common across groups, divorced wives experience a higher frequency of negative events overall (5% vs. 8%). These patterns suggest that adverse shocks, particularly related to health and employment, are more frequently observed among couples who eventually divorce. Finally, in Panel (E), we include several marriage market characteristics that capture both the size of the marriage market and the prevalence of marriage and divorce in the prefecture where the couple resides.

One may argue that satisfaction measures are indicators of marital stability on their own. Thus, we next examine the correlation between the five satisfaction measures and various characteristics of the couple. These results are reported in Figure 1, with blue cells indicating a positive correlation and red cells indicating a negative one. The heatmap highlights several consistent patterns. Differences in leisure hours between husbands and wives are negatively associated with all satisfaction measures, suggesting that imbalances in free time reduce overall marital happiness. In contrast, measures of household resources, such as family savings and the husband’s income, are positively correlated with several measures of satisfaction. The wife’s own market hours are negatively correlated with her satisfaction, while the husband’s contributions to housework show positive associations. Together, these patterns underscore the importance of both relative time allocations and financial resources in shaping women’s satisfaction with different aspects of their lives.

4.3 Sorting into Marriage

In their pioneering paper, [Becker *et al.* \(1977\)](#) identify two general causes of marital dissolution. The first is search costs, where costlier search lowers the bar for an acceptable match. In a marriage market with high search costs, one would expect a larger deviation of chosen matches from the optimal sorting equilibrium, and thus a higher likelihood of divorce. The second general cause is “surprises” or shocks that occur after marriage. Here, we indirectly examine the importance

Figure 1: Correlation between Select Covariates and Wife's Satisfaction

	Happiness	Relationship	Life	Income	Spending
Household Hours Diff. (H-W)	-0.04	0.00	0.00	-0.01	0.02
Market Hours Diff. (H-W)	0.12	0.09	0.08	0.09	0.04
Leisure Hours Diff. (H-W)	-0.09	-0.11	-0.10	-0.10	-0.07
Consumption Ratio (H/(H+W))	-0.01	-0.02	0.01	0.04	0.00
Savings Ratio (H/(H+W))	-0.01	-0.02	-0.01	0.01	0.01
Income Ratio (H/(H+W))	-0.09	-0.07	-0.07	-0.07	-0.05
Wife's Household Hours	0.09	0.04	0.03	0.01	-0.01
Wife's Market Hours	-0.10	-0.08	-0.06	-0.06	-0.03
Wife's Leisure Hours	0.00	0.03	0.03	0.05	0.04
Husband's Housework Hours	0.14	0.14	0.08	0.01	0.04
Husband's Market Hours	0.06	0.04	0.06	0.08	0.03
Husband's Leisure Hours	-0.13	-0.11	-0.10	-0.08	-0.05
Public Goods Expenditure	0.00	0.01	-0.04	-0.08	-0.04
Household Expenditure	0.00	-0.02	0.05	0.14	0.01
Husband's Savings	-0.01	-0.01	0.00	0.03	0.01
Wife's Savings	-0.03	-0.03	-0.01	0.03	0.02
Family Savings	0.15	0.13	0.19	0.24	0.16
Husband's Income	0.06	0.05	0.08	0.08	0.05
Wife's Income	-0.08	-0.06	-0.05	-0.06	-0.02

Correlation



0.2
0.1
0.0
-0.1
-0.2

Note: Across columns we list the wife's satisfaction with different aspects of her life. Satisfaction measures range from value 1–5: 1 "not at all", 2 "a little", 3 "moderately", 4 "quite much", 5 "very much". Happiness ranges from value 1–5: 1 "very unhappy", 2 "a little unhappy", 3 "neither happy nor unhappy", 4 "rather happy", 5 "very happy". Positive values indicate increased satisfaction or happiness.

of search costs and assortativeness by plotting Kaplan-Meier survival curves for spousal differences in education and age in Figure 2. In Panel (A), we classify couples into four mutually exclusive categories based on whether the husband and wife have a high (> 12 years) or low (≤ 12 years) level of education at the time of marriage. Panel (B) divides couples by spousal age differences. The vertical axis shows the probability of remaining married, while the horizontal axis plots marriage duration.

Several notable patterns emerge. We find that couples in which both spouses have high education levels are most strongly associated with marital stability, while those in which both spouses have low education levels are most strongly associated with divorce. Thus, assortative matching alone does not appear to drive marital stability; the levels of spouses' characteristics also play an important role in marital stability. This finding is consistent with evidence from previous research. For example, Raymo *et al.* (2013) use the Japanese National Fertility Survey in 2005 and the JPSC and find that educated women are less likely to get divorced within 10 years of marriage. Turning to relative age in Panel (B), we see that same-age couples (roughly 15 percent of couples in our sample) are the most stable, though the differences between categories are less stark.

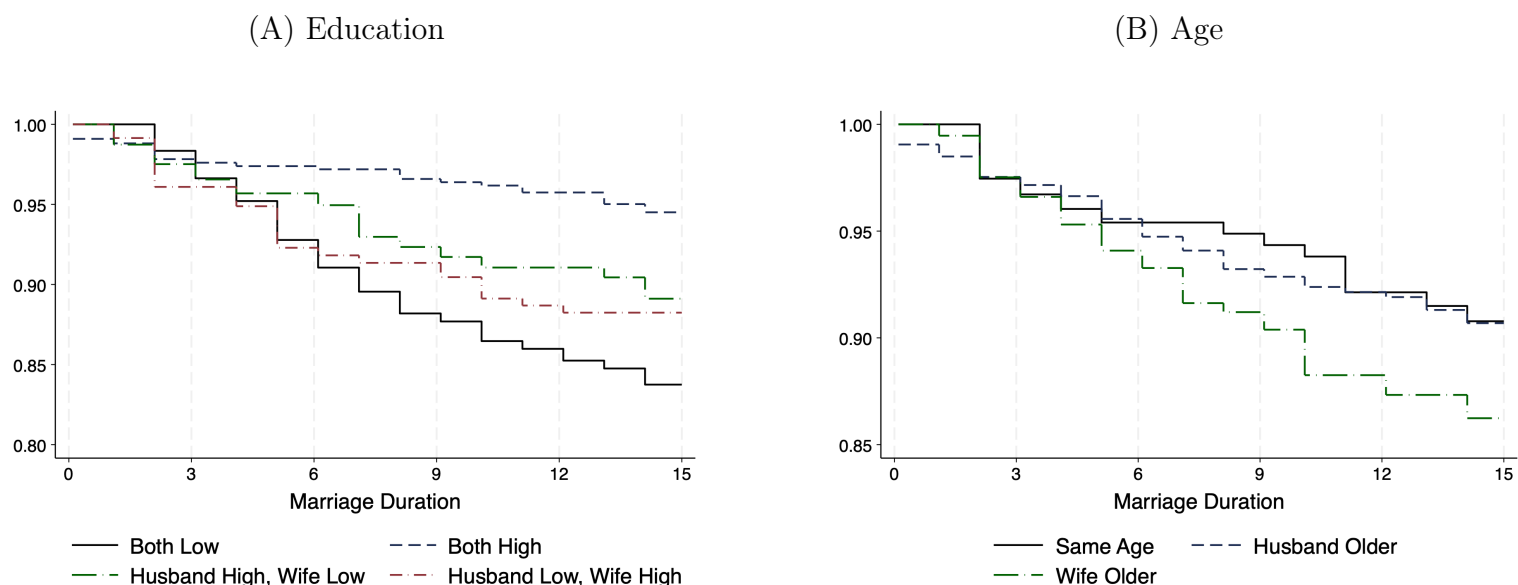


Figure 2: Kaplan-Meier Survival Curves by Spousal Differences in Education and Age

Note: Kaplan-Meier survival curves relating spousal differences in education and age to the probability of remaining married over time. The x-axis presents the duration of the marriage in years, while the y-axis shows the probability of remaining married. Panel (A) classifies couples as both having low education (12 or fewer years) or high (more than 12 years). We restrict the maximum marriage duration presented to be 15 due to few observations beyond that point.

If we compare the correlation of several couple characteristics across divorced and never-divorced couples, we see slightly higher rates of assortativeness among the never-divorced couples. The correlation of age between husbands and wives is 0.83 among couples that have yet to divorce compared to 0.77 among those that have. Similarly, we see a 0.47 correlation in years of education among the never-divorced compared to 0.37 among the divorced.

4.4 Evidence from Event Studies

We next analyze the evolution of certain labor, consumption, and savings in the years leading up to divorce. To do this, we estimate regressions with couple fixed effects that rely on within-couple variation over time.

We denote the year of divorce for couple i by T_i , and index three years prior to the divorce year (so event time runs from -3 to -1). Let y_{is}^g denote the outcome of interest (leisure hours, market work hours, domestic work hours, and share

of consumption) for an individual of gender g in couple i in year s . Based on a specification similar to Kleven *et al.* (2019) and Foerster (2021), we run the following regressions separately for men and women for the outcome variables listed above:

$$y_{is}^g = \sum_k \beta_k^g \times I[age_{is}^g = k] + \sum_z \gamma_z^g \times I[s = z] + \sum_{q=1}^3 \alpha_q^g \times I[T_i - s = q] + \psi_i + v_{is}, \quad (1)$$

where we include age dummies β_k^g for age k (first term on the right-hand side),¹³ year dummies γ_z^g (second term), and our three event time dummies α_q^g (third term) for each gender g . The omitted period is four or more years before divorce, and the indicator function $I[T_i - s = q]$ takes unity when year s is q years before divorce and zero otherwise. Couple fixed effects and error terms are denoted by ψ_i and v_{is} , respectively.

Figure 3 plots the coefficient estimates $\hat{\alpha}_q^g$ for expenditure allocations, time use measures, savings, and income management measures separately for men and women. We present estimates for how the couples' behavior in years one, two, and three prior to divorce differ from years four and earlier. Interestingly, the couples' expenditure patterns appear to change significantly in the year prior to divorce, but not earlier. Specifically, we observe a decrease in spending on public goods in Panel (A) and an increase in the wife's consumption share in Panel (B). Together, these results suggest expenditure patterns are predictive of marital stability, though this specification does not identify the direction of causation.

Moving to time use, in Panel (D) of Figure 3, we observe an increase in the wife's time in market work by nearly six hours per week relative to four or more years prior to divorce, consistent with existing research (Johnson & Skinner, 1986, 1988). We find the opposite pattern for husbands in Panels (G) and (I), as they decrease their market work by roughly 5 hours per week and increase their time spent on leisure by nearly 6 hours. The estimates indicate a clear gender asymmetry in how couples behave in the lead-up to divorce. We do not observe any change in joint or individual savings behavior (Panels (J)–(L)). Finally, we find some evidence that women controlled a larger share of household income prior to divorce, consistent with a higher consumption share for wives (Panel (N)). While these results are indicative of certain behaviors changing in the years leading up to divorce, this is not a systematic way of identifying the most important predictors. We delve deeper in Section 5 to further investigate these findings.

5 Predicting Divorce: Methods and Results

The above analyses provide suggestive evidence of the role of certain couple characteristics, such as match quality, couple behavior, and unexpected shocks, in the likelihood of divorce. We now examine these predictors jointly to identify which are most informative of divorce hazard. Guided by the theoretical framework in Section 2 and the testable implications derived in Appendix C, we focus on several key channels such as match quality, household resources and allocations, and marriage market characteristics. These categories map directly into the predictors used in our empirical models.

5.1 Cox Proportional Hazards Models

5.1.1 Model and Estimation with Delayed Entry

We begin by estimating a Cox proportional hazards model, similar to Marinescu (2016). We estimate both the standard version of that method, as well as the least absolute shrinkage and selection operator (Lasso) applied to the Cox model

¹³For household-level outcome variables, we include age dummies for both men and women.

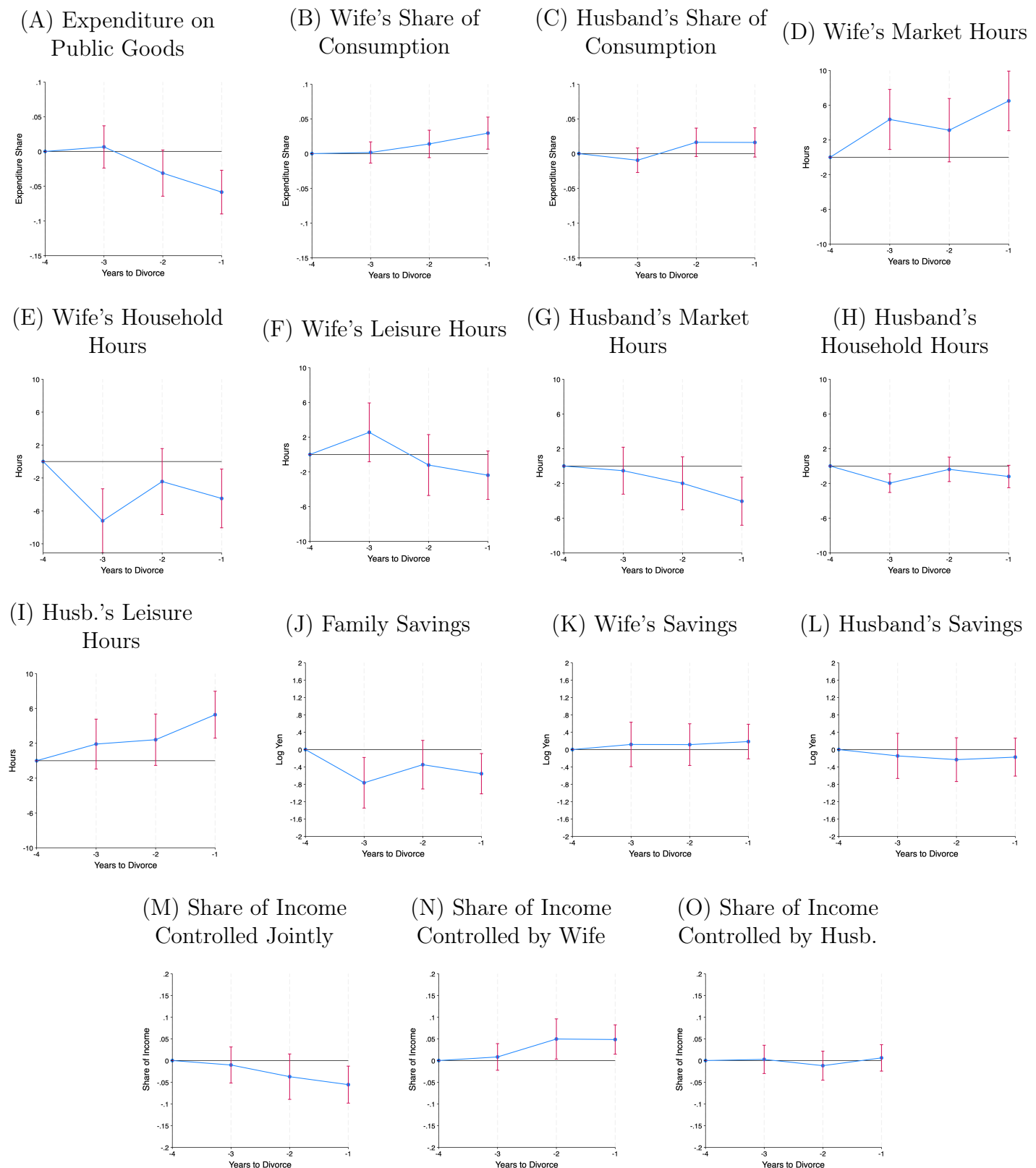


Figure 3: Evolution of Couple's Behavior Prior to Divorce

Note: The figure presents estimates on the evolution of consumption and time use measures in the lead up to divorce. We include age, year, and couple fixed effects in all specifications. We include never divorced couples to help identify age and year fixed effects. The omitted period is four or more years before divorce.

(Tibshirani, 1997). Lasso Cox adds regularization to the standard model, allowing the estimation to incorporate variable selection to reduce overfitting. This addition is especially useful in our context, where we have many predictors and multicollinearity is a concern. Without Lasso, we select a subset of all covariates based on characteristics that the past literature has identified as relevant. Therefore, we prefer to use Lasso, but we also provide additional results for the standard Cox model.

A useful feature of the Cox framework in our setting is that it can accommodate delayed entry, or left truncation, through the construction of the risk sets. Couples whose marriages began before they entered the survey do not contribute person-time prior to their first observation. Instead, they enter the risk set only once they are observed in the panel and are still married. Thus, the model compares couples who are at risk of divorce at the same marital duration, conditional on having survived to the point at which they enter the sample. This approach is valid under the standard conditional independent-truncation assumption: conditional on observed covariates and marital duration, survey entry is independent of subsequent divorce risk.

In this model, we have the following log-partial likelihood function:

$$\ln L(\boldsymbol{\psi}) = \sum_{t=1}^T \sum_{i \in D_t} \left[\mathbf{X}'_{it} \boldsymbol{\psi} - \log \left(\sum_{\ell \in R_t} \exp(\mathbf{X}'_{\ell t} \boldsymbol{\psi}) \right) \right] - \lambda \sum_{p=1}^P |\psi_p| \quad (2)$$

where $\boldsymbol{\psi}$ is the vector of regression coefficients, $\mathbf{X}_{it}$ is the covariate vector for the couple experiencing the event at time t , D_t is the set of couples who get divorced within a year, and T is the maximum observed marital duration in the estimation sample. The key adjustment for delayed entry is in the definition of the risk set R_t . Let a_i denote the marital duration at which couple i first enters the survey. Then R_t includes all couples who are observed, still married, and at risk of divorce at duration t ; equivalently, couple ℓ is included in R_t only if $a_\ell \leq t$ and the couple has not yet divorced or been censored prior to t . Thus, marriages that are already ongoing at survey entry are treated as left-truncated spells and contribute to the likelihood only after their observed entry duration. $\lambda (\geq 0)$ is the regularization parameter controlling the strength of the penalty, and P is the number of covariates. We use an 80/20 split of the data into training and testing sets. Within the training set, we use 10-fold cross-validation to select the regularization parameter.

We complement the Lasso Cox results with an unpenalized Cox proportional hazards model. Rather than allowing the penalty to determine which covariates enter the model, we specify a more parsimonious set of predictors guided by the existing literature, our theoretical framework, and the descriptive evidence. This specification also provides conventional confidence intervals for the estimated hazard ratios.

5.1.2 Lasso Cox Results

In our main specification, we include 73 predictors. The Cox model additionally allows us to interpret the direction of effects, and we can therefore use it to examine the comparative statics in Appendix C, where covariates influence the divorce hazard through their effect on the surplus from marriage. These predictors include all the variables presented in Table 2. Of the 73 predictors, Lasso Cox selects 41 with a regularization parameter of $\lambda = 0.0006$. Figure 4 presents the results in which we provide the estimated hazard ratios.

In our first model, we include all 73 predictors (represented with green diamonds). We find self-reported satisfaction measures to be important predictors. In particular, the wife's satisfaction with the relationship is among the more important

predictors of divorce, with a hazard ratio well below 1. This suggests that lower reported relationship satisfaction by the wife is associated with a higher likelihood of marital dissolution. Two of the stronger predictors of divorce are whether an important life event occurred for the wife in the previous year. Recall that these variables indicate whether the wife experienced any significant labor market changes or health issues. This result is expected, and consistent with the theory literature discussed in Section 2.

Several parental-schooling variables are also selected by the Lasso Cox model and appear to be predictive of divorce hazard. Although the mechanism is less direct than for satisfaction or recent life events, we interpret these variables as proxies for match quality and family background. Parental education may capture dimensions of the spouses' socioeconomic origins, family norms, expectations about marriage, and the degree of similarity between the spouses' family environments that are not fully reflected in the couple's own education or income. In this sense, the importance of parental schooling is consistent with the broader theoretical view that marital stability depends not only on current household resources and shocks, but also on characteristics of the match formed at the time of marriage.

In the second model, denoted by hollow circles, we exclude satisfaction variables. The motivation for these omissions is that the satisfaction measures may be more of an outcome measure correlated with our predictors of interest than a predictor. Omitting the satisfaction measures results in the Lasso selecting 40 variables, with 34 appearing in both models. Here, we find an increase in importance of several labor market variables, including differences in leisure hours. This can be interpreted as specialization within the marriage resulting in more stability.

Finally, several consumption-allocation measures are selected in both specifications. In particular, the share of expenditure devoted to public goods and the wife's and husband's private consumption shares are selected by the Lasso Cox model. The estimated hazard ratios suggest that a higher public-goods budget share is associated with a lower divorce hazard, while larger private consumption shares are associated with a higher divorce hazard. This pattern is consistent with the idea that expenditures on household public goods, including spending related to children and the family as a whole, reflect marriage-specific investments that may stabilize the relationship, whereas greater emphasis on private consumption may be associated with weaker marital stability.

5.1.3 Unpenalized Cox Results

We supplement our main results using an unpenalized Cox proportional hazards model, similar to [Marinescu \(2016\)](#). This specification has the added benefit of informing us the statistical significance of the results. The Cox proportional hazards model takes the following form (using the notation we used above), where the hazard of divorce at marriage duration t is given by $h(t, X) = h_0(t) \exp \{X\psi\}$ where $h_0(t)$ is the baseline hazard function, $X = (x_1, \dots, x_P)$ is the covariate vector, and ψ is the coefficient vector. The standard Cox model assumes that the covariates have a proportional effect on the baseline hazard.

Because we have panel data and many couple characteristics vary over marriage duration, we allow for time-dependent covariates:

$$h(t, X, Y(t)) = h_0(t) \exp \left[\sum_{p=1}^P \psi_p X_p + \sum_{m=1}^M \gamma_m Y_m(t) \right], \quad (3)$$

where ψ_p and γ_m are the coefficients corresponding to the p^{th} time-independent variable and the m^{th} time-dependent

variable, respectively. The model is estimated via partial likelihood.

Figure 5 presents the estimated hazard ratios originating from the estimation of Equation 3, which reports both the full specification and a specification that excludes the satisfaction measures. The directions and broad patterns of the estimates are generally consistent with the Lasso Cox results. The wife's satisfaction with her relationship is the clearest and most precisely estimated stabilizing correlate, with greater satisfaction associated with a substantially lower divorce hazard. Recent events experienced by the wife are associated with a higher hazard, particularly negative events, although the comparatively wide confidence interval for the latter indicates considerable uncertainty about its magnitude. Higher wife's income and the number of children aged six and over are also associated with a higher divorce hazard.

By contrast, a larger public-goods budget share is associated with a lower divorce hazard. This pattern is consistent with the interpretation of expenditures on children and other household public goods as investments in marriage-specific capital, but it should be interpreted jointly with family composition and private-consumption shares rather than as an isolated effect. The estimates are broadly stable when the satisfaction measures are omitted, suggesting that the other relationships are not driven solely by their correlation with reported marital satisfaction.

For many remaining covariates, however, the hazard ratios are close to one or their confidence intervals include one. This finding should not be interpreted as showing that these variables necessarily lack predictive content given the relatively small number of observed divorces and the strong correlation among measures of income, labor supply, consumption allocation, and family composition. Figure 5 therefore provides the strongest corroborating evidence for the roles of relationship quality and recent shocks, while offering more tentative evidence about the independent contributions of particular household-resource and allocation variables.

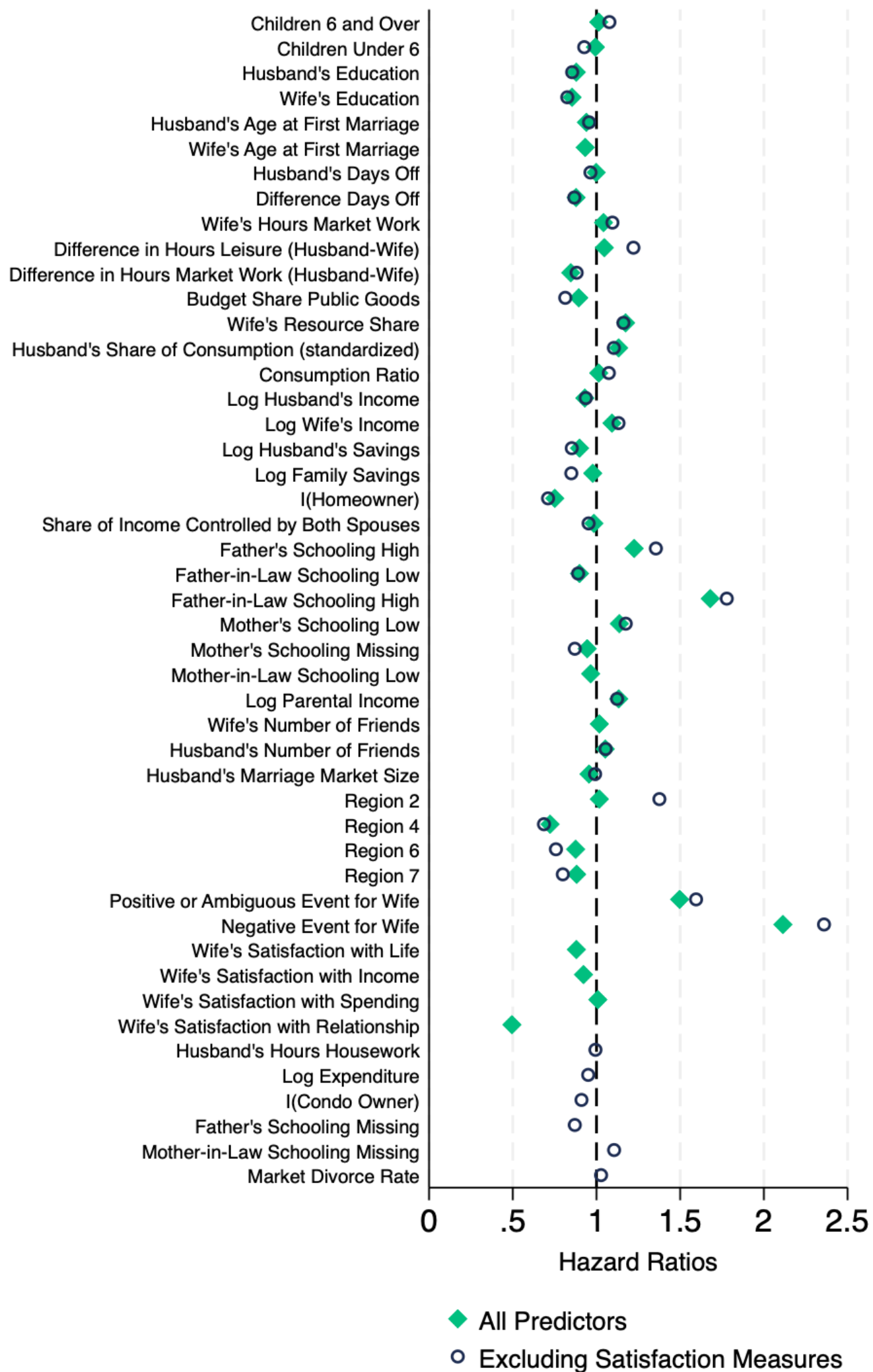


Figure 4: Cox Lasso Hazard Ratios

Note: Hazard ratios from the Lasso Cox model. Green diamonds indicate the model included all possible predictors. Black circles indicate the model excluded satisfaction measures. Continuous variables are standardized. Hazard ratios below one indicate a decrease in risk of divorce. We omit region variables for conciseness.

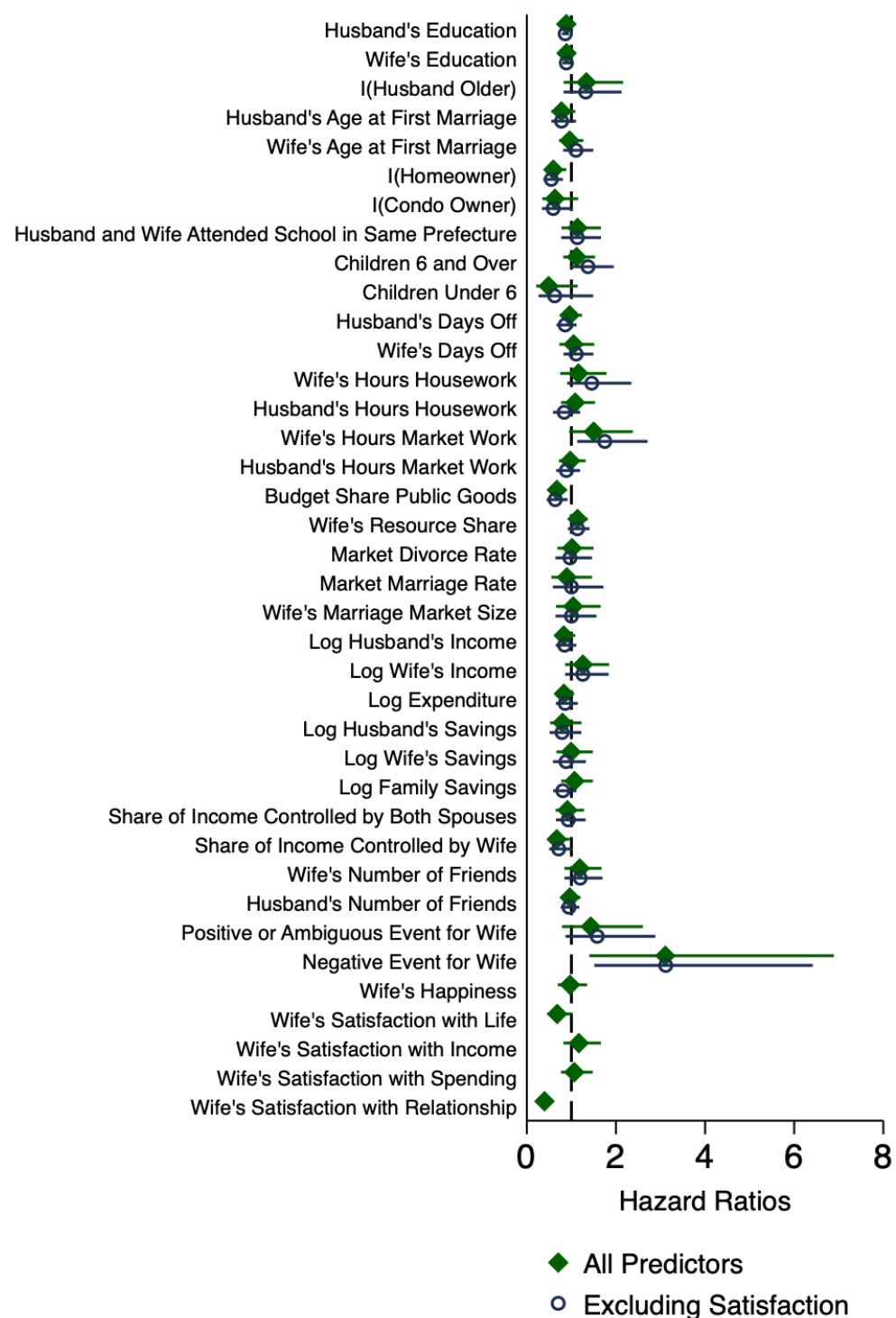


Figure 5: Cox Hazard Ratios

Note: Relative changes in the hazard rate associated with a one-unit change in each variable, holding all else fixed. Continuous variables are standardized to ease the comparison across coefficients. Region fixed effects are included in the estimation, but omitted from the figure for clarity. Hazard ratios below one indicate a decrease in risk of divorce. Horizontal lines represent 95 percent confidence intervals.

5.1.4 Sensitivity to Left Truncation

We next address concerns arising from left truncation in our sample. Because many marriages are already ongoing when respondents enter the panel, we do not observe most marriages from their onset. Consequently, the observed sample may over-represent marriages that survived long enough to be included in the survey and may not be representative of the broader population of marriages. To assess the sensitivity of our results to this issue, we re-weight the data using inverse-probability weights based on the likelihood that a marriage is observed from its start. Specifically, we estimate a logit model predicting whether a marriage is non-truncated using characteristics measured at the time of marriage, such as spouses' ages, education, and birth cohort. We then use the estimated probabilities to construct inverse-probability weights and

incorporate these weights directly into the Cox Lasso estimation.¹⁴

We present the results in Figure A8, which shows that the main Cox Lasso results are broadly robust to reweighting the sample to account for left truncation. The inverse-probability-weighted estimates preserve many of the key patterns, as the wife’s satisfaction with the relationship remains strongly negatively associated with divorce hazard, while negative events experienced by the wife remain among the strongest positive predictors. The weighted specification also continues to select several household-resource and allocation variables, including the wife’s income, the public-goods budget share, and homeownership, with signs that are generally consistent with the baseline estimates. Some individual coefficients and selected variables change under the reweighting—most notably among parental-background and regional characteristics—but there is little evidence that correcting for differential probabilities of observing marriages from their start overturns the substantive conclusions.

5.2 Random Forest

To complement the above results, we employ a random forest as an alternative method, owing to its flexibility in handling high-dimensional data with potentially complex interactions among covariates. The algorithm constructs an ensemble of decision trees, each trained on a bootstrapped sample of the data and a randomly selected subset of covariates. The final survival predictions are obtained by aggregating predictions across trees. This approach mitigates overfitting and is particularly well-suited to our empirical context, where many collinear, couple-level characteristics may jointly influence marital outcomes.

Our setting requires several deviations from the standard random forest approach. First, we are working with survival data, so our goal is to predict the time until divorce while accounting for the right-censoring of marriage duration in the data. This method, *random survival forest*, applies random forest to right-censored survival data (Ishwaran *et al.*, 2008). It is most commonly used in biomedical research, where the goal is to predict patient survival times or times until disease progression, but it has also been applied in social science settings, such as unemployment duration and credit risk (Fantazzini & Figini, 2009).¹⁵ A notable contribution related to our research question is recent work by Arpino *et al.* (2022), who applied this method to analyze predictors of divorce. However, a random survival forest does not allow for time-varying covariates. This limitation leads to our second deviation: applying recent advancements in the random survival forest literature by Devaux *et al.* (2023), which accommodate these variables. Here, we provide a high-level overview of the method. Devaux *et al.* (2023) use a competing-risk random survival forest and modify the random survival forest by incorporating longitudinal predictors using linear mixed models, which can be easily estimated using the *DynForest* package in R.

5.2.1 Notation and Estimation Process

Suppose that we observe N couples indexed by i . Let T_i be the event time (or marriage duration in our context) measured in years, with D_i being the time of divorce and t indexing years. Our analysis includes two types of variables; P time-independent covariates denoted X_{ip} for $p \in \{1, \dots, P\}$, and M continuous time-dependent covariates denoted Y_{it}^m for $m \in \{1, \dots, M\}$, which are measured for couple i at marriage duration time t . Examples of time-independent predictors include the couple’s age difference, parental education measures, and relative education at marriage. Our time-dependent predictors,

¹⁴We present the results of the model used to construct the weights in Appendix Tables A1 and A2.

¹⁵See Wang *et al.* (2019) for a survey of machine learning for survival analysis.

which comprise the majority of our features, include measures of labor supply, expenditure, savings, and satisfaction that vary over time.

The random forest procedure involves recursively splitting a bootstrapped sample of couples into two subgroups at nodes until the subgroups have reached a minimal size (3 in our main specification). At each node, the sample is split into two groups based on the predictor that maximizes the difference in divorce probability between them using the log-rank statistic. In other words, the couples are divided according to the predictor that provides the strongest discrimination between the two groups at that node.

The main modification of [Devaux *et al.* \(2023\)](#) from the existing literature is to incorporate covariate dynamics. At each node, the trajectory of time-dependent covariates is specified using a mixed model.¹⁶ Maximum likelihood estimation of the mixed model is computed at each node. Specifically, at each node, the dynamics of each time-dependent predictor m is collapsed into a set of time-independent features using the following mixed model:

$$Y_{it}^m = Z^m(t)\beta^m + Z^m(t)b_i^m + \epsilon_{it}^m \quad (4)$$

where Y_{it}^m is the value of the covariate m of couple i at marriage duration t , and Z^m is the vector of functions of time associated with the fixed and random effects of the mixed model.¹⁷ b_i^m and ϵ_{it}^m are assumed to be mean-zero normal random errors. We assume that the trajectory of the covariates in the mixed model is quadratic in time. We experimented with alternative specifications (e.g., including a cubic term), but the results were largely unchanged. We then use the predicted random effects of the mixed model as time-independent features to be included as candidate features in the random forest.¹⁸

To summarize, the tree building process proceeds in the following steps:

1. A pre-specified number of time-independent and time-dependent candidate predictors are randomly selected.
2. The time-dependent predictors are converted into time-independent predictors using a mixed model. The mixed model is estimated using repeated observations from the individual couples.
3. For each candidate feature, we generate potential splits, and quantify the distance between the resulting two groups. Features and split points are chosen to maximize the value of a log-rank statistic.
4. The above steps are repeated until the stopping criteria are met.¹⁹

Finally, because we do not observe complete marital histories for all couples, left truncation remains a concern. This issue is more consequential for DynForest than for the Cox specifications, since DynForest does not explicitly accommodate delayed entry. We therefore apply the same inverse-probability weights used in the earlier analysis to all random-forest estimates. The weights are incorporated through subject-level resampling conducted at the couple level, rather than the couple-year level, thereby preserving each couple's complete observed longitudinal trajectory.

Each couple is assigned an inverse-probability weight, and we draw a new training sample with replacement, with selection probabilities proportional to these weights. We then estimate the DynForest model on this IPW-resampled pseudo-

¹⁶The estimation program used in our analysis currently can incorporate only continuous time-varying variables, though discrete ones can theoretically be added.

¹⁷The mixed model is estimated only with couples present at a given node.

¹⁸More specifically, [Devaux *et al.* \(2023\)](#) convert time-dependent features to time-independent ones as the predicted couple-level deviations to the mean trajectory.

¹⁹The stopping criteria are based on the minimal number of couples in a node, or the number of divorces required to split the node.

population using the same predictors and tuning parameters as in the baseline specification. This procedure approximates a weighted random forest by increasing the influence of couples that represent a larger share of the target population under the inverse-probability weighting scheme.

5.2.2 DynForest Results

We begin with the random forest results.²⁰ Figure A4 in the Appendix reports variable importance (VIMP), measured by the increase in out-of-bag prediction error when a predictor is permuted. Higher values indicate greater predictive importance. Because individual VIMP conditions on all remaining predictors, however, it can understate the importance of variables whose information is shared with closely correlated covariates. We therefore treat these results as suggestive and later use Group VIMP, which jointly permutes related variables and does not suffer from this collinearity problem, as the preferred estimates.

Panel (A) of Figure A4 in the Appendix presents the variable-importance results for characteristics measured at or near the time of marriage, including parental characteristics, education, work experience, homeownership, and the spouses' ages. Several of these match-quality variables have meaningful predictive content once incorporated into the random forest. The wife's work experience at marriage has the largest VIMP in this group, followed by the father's education, the husband's age at marriage, and the husband's work experience. The wife's age at marriage, parental income, and several other parental-education measures also contribute to prediction. Homeownership at marriage (or survey entry for couples already married when first observed) is predictive, but is not among the leading variables in the weighted specification. By contrast, the husband's education and condominium ownership contribute little or negatively to predictive performance.

These results complement the patterns suggested by the Kaplan-Meier curves in Figure 2. While the Kaplan-Meier analysis highlights differences in marital stability by spouses' education and age, the random-forest results indicate that predictive content is distributed across a broader set of characteristics measured at or near the start of marriage. In particular, work experience, parental background, and spouses' ages remain informative even after accounting for time-varying household characteristics. This suggests that characteristics established near the beginning of the marriage contain information about subsequent divorce risk, although no single dimension of match quality dominates across all measures.

Panel (B) shows that marriage characteristics related to the wife's income are among the most important predictors of divorce. The wife's log income has a VIMP of nearly 1 percent, meaning that permuting its values increases the out-of-bag (OOB) error by that amount. Other closely related predictors, such as the difference in hours worked between spouses, also rank highly. Wife's income may proxy not only for direct financial resources but also for intra-household bargaining power. It may further reflect unobserved traits such as job stability or social status, both of which are likely to influence marital stability.

Children in the household also have predictive content, though their importance is more modest than that of several income and time-use variables. The number of children aged six and over has a positive VIMP, while the number of children under six is also predictive but smaller in magnitude. These results are consistent with the idea that family composition is informative about marital stability. Indeed, the Lasso Cox results in Figure 4 suggest that having older children is

²⁰We summarize different aspects of the random forest model in the Appendix. Figure A3 illustrates how we determined different values of the hyperparameters used in our model by measuring how the DynForest out-of-bag prediction error varies across different tuning-parameter combinations, including the number of candidate variables, node size, and the minimum split. The lowest observed OOB error occurs when the model uses 12 candidate variables, a node size of 3, and a minimum split of 2. The model performance is sensitive to tuning choices, with smaller node-size and minimum-split settings generally producing lower prediction error. In all random forest specifications, we use 1000 trees.

associated with an increase in the divorce hazard, which cautions against interpreting children solely as marriage-specific capital that stabilizes the marriage. One of the most novel aspects of the JPSC data is its inclusion of individual-level consumption measures. In the random-forest results, however, individual consumption shares and the share of expenditure devoted to public goods have limited predictive importance when considered separately. One possible explanation is that intra-household allocation variables change sharply only as divorce approaches, as Figure 3 suggests. DynForest models time-varying covariates using linear mixed models, so variables that exhibit abrupt, one-time changes may have limited predictive importance. By contrast, Cox Lasso is better suited to capturing such short-run changes and identifies the public-goods consumption share as an important predictor. Taken together, the two approaches capture complementary dimensions of divorce risk.

In Panels (C) and (D) of Figure A4, we present the VIMP of the stated satisfaction measures and marriage-market characteristics, respectively. Among the satisfaction measures, the wife’s satisfaction with her relationship is the most predictive, followed by her overall happiness and satisfaction with life. The marriage-market variables have more limited predictive content: local marriage and divorce rates have positive VIMP values, while the husband’s marriage-market size contributes little or negatively. Thus, although these results are broadly consistent with models in which subjective marital quality and outside options affect divorce (Lundberg & Pollak, 1993), the baseline VIMP results suggest that marriage-market conditions are less predictive than several match-quality and household-resource measures.

A drawback of VIMP in our context is that many predictors are highly correlated. For example, by construction, men’s and women’s shares of total consumption and the share of public-good expenditures sum to one, so permuting one variable while holding the others fixed can generate unrealistic variation. Our preferred approach to address this issue is to use gVIMP, which measures the importance of sets of variables, as shown in Figure 6. We consider two categorizations. First, we group variables according to the panels in Figure A4: match quality, household resources and allocations, satisfaction measures, and marriage market characteristics. We then disaggregate these categories to compare narrower groups, such as expenditure allocations, time use, income and savings, and parental characteristics.

In the aggregate grouping, match quality is the most predictive category, followed by household resources and allocations, satisfaction measures, and marriage-market characteristics. The more detailed grouping shows that this pattern is driven primarily by time use, followed by parental characteristics and income and savings. Satisfaction measures retain predictive content, but they are less important than these leading groups, while marriage-market characteristics and expenditure allocations play more modest roles. These results indicate that the strongest group-level predictors are characteristics measured at or near marriage and time-varying household resources, rather than satisfaction or marriage-market conditions alone.

An alternative importance measure is minimal depth. Variables that are more important tend to appear both more frequently and earlier in the tree-building process. To compute this measure, we follow Devaux *et al.* (2023) and modify the main specification so that all variables are available for selection in each tree. We estimate this model using 2000 trees and plot the distribution of each variable’s minimal depth across trees.

The results are presented as box plots in Figure A5 in the Appendix. For each time-varying predictor, we report separately the random intercept, linear slope, and quadratic term from the mixed model, denoted by `.bi0`, `.bi1`, and `.bi2`, respectively. The number on the right-hand side indicates how many of the 2,000 trees contain the corresponding

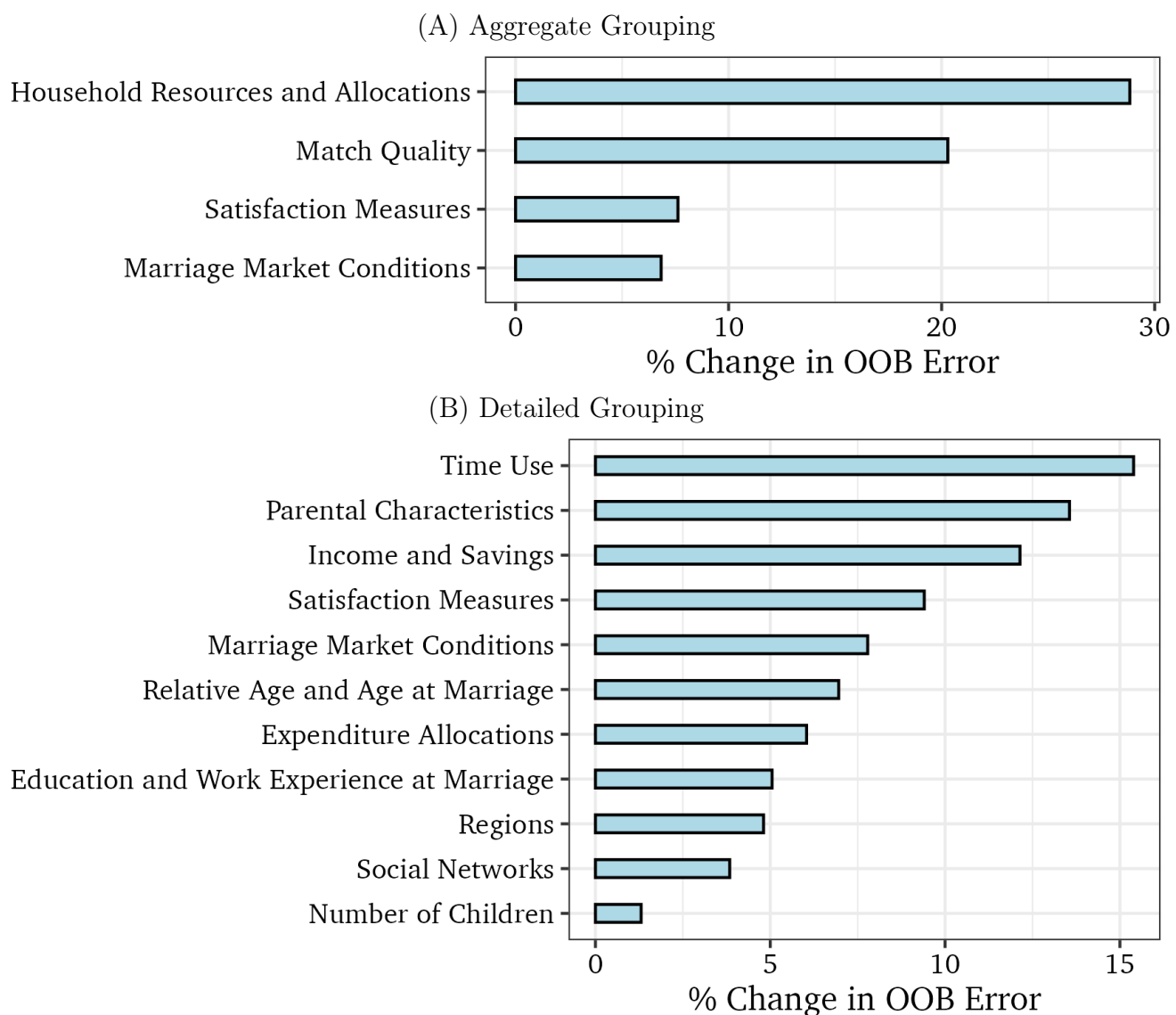


Figure 6: Group Variable Importance

Note: Group Variable Importance (gVIMP) is displayed as the percentage loss in the Out-of-Bag Error. In panel (A), we group variables following the sub-figures of Figure A4. In panel (B) we disaggregate match quality and household resources and allocations into smaller categories of related variables. Income and savings include log individual incomes of both spouses, log individual and family savings, and relative income and savings. Expenditure allocations include all consumption variables, and the shares of income controlled by the wife and jointly by the couple. Time use includes hours spent in market work, household work, and leisure, as well as differences in time spent in the categories across spouses. Time use also includes days off per week for both spouses. Social networks consist of the number of male and female friends for the wife. The remaining categories are self-explanatory. Estimates are obtained from an IPW-resampled DynForest model, where couples are sampled with replacement in proportion to their inverse-probability weights and each selected couple's full longitudinal history is retained.

feature. Features are considered more predictive when they appear in more trees and have lower minimal depth. Several time-fixed characteristics appear especially frequently, including the spouses' ages and work experience at marriage, relative education, parental education, whether the spouses came from the same prefecture, and some regional indicators. Among the time-varying predictors, the random-intercept components of the difference in spouses' days off, the income ratio, log expenditure, and the local divorce rate tend to appear frequently and relatively early. The wife's happiness and satisfaction measures—particularly satisfaction with the relationship—also have comparatively low minimal depths across multiple trajectory components. Overall, the figure suggests that predictive information is distributed across characteristics fixed at or near marriage, household resources and time use, local conditions, and subjective well-being; it also shows that the different trajectory components of a given longitudinal predictor can differ substantially in their importance.

We make several additional variable restrictions to further investigate which variables are important. First, self-reported satisfaction could be just a proxy for marital stability as opposed to a determinant. Thus, if time use was a determinant of satisfaction with one's partner, including satisfaction measures in the analysis may subsume the importance of the time use variables. As a result, we omit any self-reported satisfaction measure from the analysis and re-estimate the model. We report these results in Figure A6 in the Appendix.

Comparing Figure A6 with Figure A4, the overall ranking of predictors is broadly stable after satisfaction measures are omitted. The clearest changes occur among labor-supply and income variables in panel (B): the importance of the wife's market-work hours increases, as does the importance of the spouses' income ratio and the difference in their days off. These results suggest that the predictive content otherwise captured by satisfaction measures overlaps primarily with couples' market-work and relative-income arrangements, rather than with the presence of young children or the wife's domestic labor.

A similar pattern emerges at the group level when comparing Figure 6 with Figure A7. Omitting the satisfaction measures leaves the broad ranking of the remaining categories largely unchanged. Within the detailed grouping, time use continues to have the greatest predictive importance and becomes even more prominent once satisfaction is excluded. Income and savings also remain among the leading groups and move ahead of parental characteristics, whereas expenditure allocations, family composition, and social networks change relatively little. Thus, the information contained in the satisfaction measures appears to overlap most strongly with observed differences in couples' time allocation and economic circumstances, rather than with consumption allocations or marriage-market conditions.

5.3 Additional Analyses

5.3.1 Cohort Effects

Next, we examine changes in predictor importance by cohort. We are concerned that certain characteristics of couples may be more or less relevant today than they were in the past. To investigate this, we estimate a Cox proportional hazards model where we interact selected covariates with a cohort dummy. Specifically, we compare wives born in the 1960s to those born in later cohorts and plot the interaction terms as ratios of hazard ratios in Figure A9. Values above one indicate that the association between a predictor and divorce hazard is stronger for the 1960s cohort relative to later cohorts, while values below one indicate a weaker association.

Overall, Figure A9 provides limited evidence of systematic cohort heterogeneity. Most interaction estimates are close to

one and their confidence intervals include one, suggesting that the main predictors of divorce operate similarly across cohorts. This is particularly true for several core household-resource and time-use variables, including husbands' and wives' income, savings, housework, and market work. Thus, although family roles and marriage-market conditions have changed across cohorts in Japan, the predictive relationships documented in our baseline models do not appear to be driven exclusively by older cohorts.

There are, however, several suggestive patterns. The interaction for the public-goods budget share is above one, indicating that the association between public-goods expenditure and marital stability may be weaker for the 1960s cohort than for later cohorts. This is consistent with the possibility that shared household expenditures, including spending on children and other household public goods, have become more informative of marital stability among more recent cohorts. Similarly, the point estimate for total household expenditure is above one, although the confidence interval is wide, indicating that differences in expenditure-related predictors across cohorts should be interpreted cautiously.

Some marriage-market and family-composition variables also display modest cohort differences. The interaction for the husband's marriage-market size is above one, while the interactions for children, especially children aged six and over, are below one. These patterns suggest that outside-option and child-related mechanisms may vary across cohorts, but the estimates are imprecise and do not provide strong evidence of statistically meaningful differences. The interaction for the wife's satisfaction with the relationship is also below one, suggesting that relationship satisfaction may be at least as predictive, and possibly more predictive, for later cohorts than for the 1960s cohort.

5.3.2 Marriage Market Conditions and Divorce

We expect marriage-market conditions to be important predictors of divorce, with outside options playing a particularly important role when match quality is low, as formalized in Prediction 4 of the model presented in Appendix Section C. In the model, improved marriage-market conditions raise a spouse's outside option, but this increase is most likely to translate into divorce when the value of remaining married is already low. Since match quality is not directly observed, we use the wife's reported satisfaction measures as proxies for the quality of the match. The empirical prediction is therefore that favorable marriage-market conditions should be more strongly associated with divorce among women reporting lower satisfaction. Equivalently, when satisfaction is coded so that higher values indicate better match quality, the interaction between marriage-market conditions and satisfaction should generally have a hazard ratio below one.

Figure A10 presents Cox hazard ratios from a specification that interacts marriage-market characteristics with the wife's satisfaction measures. The results provide limited support for Prediction 4. The wife's satisfaction with the relationship remains negatively associated with divorce hazard, consistent with the broader finding that lower match quality predicts marital instability. However, the interaction terms between marriage-market conditions and satisfaction are mostly close to one and do not show a clear, systematic pattern. In particular, several interactions involving marriage-market size and satisfaction do not have hazard ratios below one, and some are above one, suggesting that stronger outside options do not appear to be especially destabilizing only among less-satisfied couples. Overall, Figure A10 suggests that relationship satisfaction is an important predictor of divorce, but it provides weak evidence that marriage-market conditions have a larger effect when reported match quality is low. This prediction may not hold if observed marriage-market measures are too coarse to capture each spouse's true outside option. Indeed, consistent with the main specifications where marriage-market

conditions have limited predictive content, divorce decisions may depend more on other observables.

5.4 Consistency with empirical predictions

Let us now compare the results presented above with the predictions offered in Section 2. First, marriage market conditions as a group have a modest predictive power for divorce (Figure 6). We also see from Figure A4 that divorce rate is a better predictor than market sizes, which is not surprising as the market-size measures used in this study are just crude proxies. As these findings suggest that the outside options may affect the divorce risk, our empirical results are consistent with Prediction 1.

Second, satisfaction measures are good predictors of divorce. It is notable that wife’s happiness and satisfaction with her relationship are particularly important predictors (Figure A4). It is also notable that the levels of these predictors tend to be frequently chosen and have a low minimum depth (Figure A5), indicating that they are important regardless of the selection of other predictors. Wife’s satisfaction with the relationship is also an important predictor of divorce hazard in the Cox model (Figures 4 and 5). These indicate that our results broadly support Prediction 2. While Prediction 2 is concerned with the effect of the shock to the marriage, we would also expect that the presence of larger marriage-specific capital leads to a lower divorce hazard by a similar logic. We also see that match quality variables are important predictors of divorce.

Third, Figure 3(A) shows a decline in public-goods expenditure in the period immediately preceding divorce. In addition, Section 5 shows that consumption allocation variables, including public-good expenditure shares, are important predictors in the random forest. Further, as Figure 4 shows, the divorce hazard for the budget share of public goods is considerably smaller than unity. These observations are consistent with Prediction 3. Finally, as discussed in the previous subsection, we have weak evidence for Prediction 4.

6 Conclusion

This paper provides new empirical insights into the predictors of divorce. We leverage a unique panel dataset and combine traditional econometric approaches with machine learning techniques to identify characteristics associated with divorce risk. Our findings indicate that divorce is predicted by a combination of factors, including measures of match quality, time-varying household resources, labor supply, satisfaction, and marriage-market conditions. In the baseline group variable-importance results, match-quality measures and household resources are especially predictive, while consumption allocation and marriage-market variables play more modest roles. The predictive patterns are broadly consistent with the theoretical mechanisms outlined in Appendix C, particularly the relevance of match quality, household resources, intra-household allocation, and outside options to marital stability.

We make several contributions to the literature on marital stability and divorce prediction. First, we apply a novel machine learning algorithm, DynForest, to model divorce as a time-to-event outcome using panel data with time-varying covariates. This extends the existing empirical literature by allowing for a richer analysis beyond what traditional survival models or static random forest methods can accommodate.

Second, we incorporate detailed data on intra-household consumption allocations, a dimension rarely available in existing

datasets. By analyzing how expenditures for private and public goods evolve and relate to divorce, we provide new evidence on the role of consumption inequality within households. Group-level variable importance measures further show that, while individual consumption variables may appear only modestly predictive of divorce on their own, their joint contribution matters. Third, we study Japan, a country that has undergone notable demographic and social transitions in family formation and dissolution. By examining Japan, where divorce norms are shifting and marriage rates are declining, we add evidence from an important institutional and cultural context that has been understudied in the empirical divorce literature.

Although the machine learning methods employed here are not suited to establish causality, they offer strong predictive validity and guide future empirical and structural work. Future research could examine whether changes in intra-household allocations or policy interventions, such as childcare subsidies or labor market reforms, causally affect marital stability.

References

- Abdel-Sater, Raed. 2022. Marriage dissolution in the United States: A survival analysis approach. *Journal of Divorce & Remarriage*, **63**(4), 235–261. [5]
- Amato, Paul R. 2010. Research on divorce: Continuing trends and new developments. *Journal of Marriage and Family*, **72**(3), 650–666. [2]
- Arpino, Bruno, Le Moglie, Marco, & Mencarini, Letizia. 2022. What tears couples apart: A machine learning analysis of union dissolution in Germany. *Demography*, **59**(1), 161–186. [2], [3], [4], [25]
- Becker, Gary S. 1974. A theory of marriage: Part II. *Journal of Political Economy*, **82**(2, Part 2), S11–S26. [4], [5], [7], [10]
- Becker, Gary S, Landes, Elisabeth M, & Michael, Robert T. 1977. An economic analysis of marital instability. *Journal of Political Economy*, **85**(6), 1141–1187. [2], [4], [5], [7], [15]
- Becker, GS. 1981. *A Treatise on the Family* Cambridge, MA: Harvard Univ. [2]
- Booth, Alan, Johnson, David, & Edwards, John N. 1983. Measuring marital instability. *Journal of Marriage and Family*, **45**(2), 387–394. [2]
- Charles, Kerwin Kofi, & Stephens, Jr, Melvin. 2004. Job displacement, disability, and divorce. *Journal of Labor Economics*, **22**(2), 489–522. [2], [5]
- Cherchye, Laurens, Demuynck, Thomas, De Rock, Bram, & Vermeulen, Frederic. 2017. Household consumption when the marriage is stable. *American Economic Review*, **107**(6), 1507–1534. [10]
- Cherchye, Laurens, De Rock, Bram, Vermeulen, Frederic, & Walther, Selma. 2021. Where did it go wrong? Marriage and divorce in Malawi. *Quantitative Economics*, **12**(2), 505–545. [5]
- Cherchye, Laurens, Samang, Helene Van, & Vermeulen, Frederic. 2025 (June). *From Vows to Breakups: A Structural Approach to Divorce in the US*. Discussion Paper DPS 25.12. KU Leuven, Faculty of Economics and Business. [5]

- Cherlin, Andrew. 1977. The effect of children on marital dissolution. *Demography*, **14**, 265–272. [5]
- Chiappori, Pierre-André. 1988. Rational Household Labor Supply. *Econometrica*, 63–90. [5]
- Chiappori, Pierre-André. 1992. Collective Labor Supply and Welfare. *Journal of Political Economy*, 437–467. [5]
- Chiappori, Pierre-Andre, Fortin, Bernard, & Lacroix, Guy. 2002. Marriage market, divorce legislation, and household labor supply. *Journal of Political Economy*, **110**(1), 37–72. [6]
- Chiappori, Pierre-André, Iyigun, Murat, & Weiss, Yoram. 2015. The Becker–Coase theorem reconsidered. *Journal of Demographic Economics*, **81**(2), 157–177. [5]
- Chiappori, Pierre-André, Radchenko, Natalia, & Salanié, Bernard. 2018. Divorce and the duality of marital payoff. *Review of Economics of the Household*, **16**, 833–858. [2]
- Dahl, Gordon B, & Moretti, Enrico. 2008. The demand for sons. *Review of Economic Studies*, **75**(4), 1085–1120. [5]
- Devaux, Anthony, Helmer, Catherine, Genuer, Robin, & Proust-Lima, Cécile. 2023. Random survival forests with multivariate longitudinal endogenous covariates. *Statistical Methods in Medical Research*, **32**(12), 2331–2346. PMID: 37886845. [3], [4], [25], [26], [28]
- Fantazzini, Dean, & Figini, Silvia. 2009. Random survival forests models for SME credit risk measurement. *Methodology and Computing in Applied Probability*, **11**, 29–45. [25]
- Farnham, Martin, Schmidt, Lucie, & Sevak, Purvi. 2011. House prices and marital stability. *American Economic Review*, **101**(3), 615–619. [5]
- Foerster, Hanno. 2021. *Untying the Knot: How Child Support and Alimony Affect Couples’ Decisions and Welfare*. Boston College. [18]
- Fukuda, S., Raymo, J.M., & Yoda, S. 2020. Revisiting the educational gradient in marriage in Japan. *Journal of Marriage and Family*, **82**, 1378–1396. [9]
- Gardner, Jonathan, & Oswald, Andrew J. 2006. Do divorcing couples become happier by breaking up? *Journal of the Royal Statistical Society Series A: Statistics in Society*, **169**(2), 319–336. [6]
- Güven, Cahit, Senik, Claudia, & Stichnoth, Holger. 2012. You can’t be happier than your wife. Happiness gaps and divorce. *Journal of Economic Behavior & Organization*, **82**(1), 110–130. [2], [6]
- Heaton, Tim B. 1990. Marital stability throughout the child-rearing years. *Demography*, **27**, 55–63. [5]
- Ishwaran, Hemant, Kogalur, Udaya B, Blackstone, Eugene H, & Lauer, Michael S. 2008. Random survival forests. *Annals of Applied Statistics*, **2**(3), 841–860. [3], [25]
- Iwasawa, M. 2008. Recent trends in first marriage and divorce and their effects on fertility change in Japan (in Japanese). *Journal of Population Problems (Jinko Mondai Kenkyu)*, **64**(4). [8]

- Johnson, William R, & Skinner, Jonathan. 1986. Labor supply and marital separation. *American Economic Review*, 455–469. [18]
- Johnson, William R, & Skinner, Jonathan. 1988. Accounting for changes in the labor supply of recently divorced women. *Journal of Human Resources*, 417–436. [18]
- Kabátek, Jan, & Ribar, David C. 2021. Daughters and divorce. *Economic Journal*, **131**(637), 2144–2170. [5]
- Kalmijn, Matthijs. 1999. Father involvement in childrearing and the perceived stability of marriage. *Journal of Marriage and the Family*, 409–421. [5]
- Kalmijn, Matthijs, & Poortman, Anne-Rigt. 2006. His or her divorce? The gendered nature of divorce and its determinants. *European Sociological Review*, **22**(2), 201–214. [5]
- Katsurada, K. 1997. Application of social theories to explain the low divorce rate in Japan compared to the U.S. *Comparative culture : the Journal of Miyazaki International College*, **3**, 105–114. [9]
- Kaufman, G., & Taniguchi, H. 2010. Marriage and happiness in Japan and the United States. *International Journal of Sociology of the Family*, **36**(1), 25–48. [8]
- Kleven, Henrik, Landais, Camille, & Søgaaard, Jakob Egholt. 2019. Children and gender inequality: Evidence from Denmark. *American Economic Journal: Applied Economics*, **11**(4), 181–209. [18]
- Kumagai, F. 1983. Changing Divorce in Japan. *Journal of Family History*, **8**(1). [8]
- Kumagai, F. 2015. *Family Issues on Marriage, Divorce, and Older Adults in Japan*. Springer Singapore. [8]
- Kureishi, Wataru, Paule-Paludkiewicz, Hannah, Tsujiyama, Hitoshi, & Wakabayashi, Midori. 2025. *Stuck in a Marriage: Labor Market Shocks, Divorce and Intra-Household Reallocation*. Tech. rept. 01/2025. Deutsche Bundesbank Discussion Paper. [2], [4]
- Lafortune, Jeanne, & Low, Corinne. 2017. Tying the double-knot: The role of assets in marriage commitment. *American Economic Review*, **107**(5), 163–167. [5]
- Lafortune, Jeanne, & Low, Corinne. 2023. Collateralized marriage. *American Economic Journal: Applied Economics*, **15**(4), 252–291. [5]
- Lee, Kristen Schultz, Tufiş, Paula A., Alwin, Duane F., & Teachman, Jay. 2010. Separate spheres or increasing equality? Changing gender beliefs in postwar Japan. *Journal of Marriage and Family*, **72**, 184–201. [9]
- Lee, Wang-Sheng, & McKinnish, Terra. 2018. The marital satisfaction of differently aged couples. *Journal of Population Economics*, **31**(2), 337–362. [5]
- Lehrer, Evelyn L, & Chiswick, Carmel U. 1993. Religion as a determinant of marital stability. *Demography*, **30**, 385–404. [5]

- Lewbel, Arthur. 2019. The identification zoo: Meanings of identification in econometrics. *Journal of Economic Literature*, **57**(4), 835–903. [4]
- Lim, S., & Raymo, J.M. 2016. Marriage and Women’s Health in Japan. *Journal of Marriage and Family*, **78**, 780–796. [8]
- Lise, Jeremy, & Yamada, Ken. 2019. Household sharing and commitment: Evidence from panel data on individual expenditures and time use. *Review of Economic Studies*, **86**(5), 2184–2219. [2]
- Lundberg, S., & Pollak, R.A. 2003. Efficiency in Marriage. *Review of Economics of the Household*, **1**, 153–167. [6]
- Lundberg, Shelly. 2012. Personality and marital surplus. *IZA Journal of Labor Economics*, **1**, 1–21. [2], [6]
- Lundberg, Shelly, & Pollak, Robert A. 1993. Separate spheres bargaining and the marriage market. *Journal of Political Economy*, **101**(6), 988–1010. [5], [28]
- Lyngstad, Torkild Hovde, & Jalovaara, Marika. 2010. A review of the antecedents of union dissolution. *Demographic research*, **23**, 257–292. [5]
- Manser, M., & Brown, M. 1980. Marriage and household decision-making: A bargaining analysis. *International Economic Review*, **21**(1), 31–44. [5]
- Marinescu, Ioana. 2016. Divorce: what does learning have to do with it? *Labour Economics*, **38**, 90–105. [5], [18], [21]
- Mazzocco, Maurizio. 2007. Household intertemporal behaviour: A collective characterization and a test of commitment. *Review of Economic Studies*, **74**(3), 857–895. [6]
- Mazzocco, Maurizio, Ruiz, Claudia, & Yamaguchi, Shintaro. 2014. Labor supply and household dynamics. *American Economic Review*, **104**(5), 354–359. [6]
- McElroy, M. B., & Horney, M. J. 1981. Nash-bargained household decisions: Toward a generalization of the theory of demand. *International Economic Review*, **22**(2), 333–349. [5]
- Nishida, S., Kimura, M., & Hayashi, K. 1987. Marriage rate, divorce rate and birth rate in Japan for the period between 1872 and 1898: A Re-examination (in Japanese). *Japanese Journal of Health and Human Ecology (Minzoku Eisei)*, **53**(4), 184–191. [8]
- Olafsson, Arna, & Steingrimsdottir, Herdis. 2020. How does daddy at home affect marital stability? *Economic Journal*, **130**(629), 1471–1500. [5]
- Peters, H Elizabeth. 1986. Marriage and divorce: Informational constraints and private contracting. *American Economic Review*, **76**(3), 437–454. [2]
- Piao, Xiangdan. 2021. Marriage stability and private versus shared expenditures within families: Evidence from Japanese families. *Social Indicators Research*, **153**(2), 533–559. [3], [4]
- Rainer, Helmut, & Smith, Ian. 2010. Staying together for the sake of the home?: House price shocks and partnership dissolution in the UK. *Journal of the Royal Statistical Society Series A: Statistics in Society*, **173**(3), 557–574. [5]

- Raymo, James M, Fukuda, Setsuya, & Iwasawa, Miho. 2013. Educational differences in divorce in Japan. *Demographic Research*, **28**, 177–206. [17]
- Retherford, R.D., Ogawa, N., & Matsukura, R. 2001. Later marriage and less marriage in Japan. *Population and Development Review*, **27**(1), 65–102. [8]
- Rosenfeld, Michael J. 2018. Who wants the breakup? Gender and breakup in heterosexual couples. *Pages 221–243 of: Alwin, Duane F., Felmlee, Diane H., & Kreager, Derek A. (eds), Social networks and the life course: Integrating the development of human lives and social relational networks*. Springer. [5]
- Rosenfeld, Michael J, & Roesler, Katharina. 2024. Stability and change in predictors of marital dissolution in the US 1950–2017. *Journal of Marriage and Family*, **86**(1), 154–175. [5]
- Seaman, Shaun R, & White, Ian R. 2013. Review of inverse probability weighting for dealing with missing data. *Statistical methods in medical research*, **22**(3), 278–295. [12]
- South, Scott J, & Lloyd, Kim M. 1995. Spousal alternatives and marital dissolution. *American Sociological Review*, 21–35. [6]
- Stevenson, Betsey, & Wolfers, Justin. 2007. Marriage and divorce: Changes and their driving forces. *Journal of Economic Perspectives*, **21**(2), 27–52. [10]
- Svarer, Michael, & Verner, Mette. 2008. Do children stabilize relationships in Denmark? *Journal of Population Economics*, **21**, 395–417. [5]
- Tibshirani, Robert. 1997. The lasso method for variable selection in the Cox model. *Statistics in Medicine*, **16**(4), 385–395. [20]
- Tokuhiro, Y. 2009. *Marriage in Contemporary Japan*. Routledge. [8]
- Velilla, Jorge, Molina, José Alberto, & Chiappori, Pierre. 2025. *The price of breaking up: Wage shocks and household dissolution*. Boston College Working Papers in Economics 1096. Boston College Department of Economics. [5]
- Voena, Alessandra. 2015. Yours, mine, and ours: Do divorce laws affect the intertemporal behavior of married couples? *American Economic Review*, **105**(8), 2295–2332. [6]
- Waite, Linda J, & Lillard, Lee A. 1991. Children and marital disruption. *American Journal of Sociology*, **96**(4), 930–953. [5]
- Wang, Ping, Li, Yan, & Reddy, Chandan K. 2019. Machine learning for survival analysis: A survey. *ACM Computing Surveys (CSUR)*, **51**(6), 1–36. [3], [25]
- Weiss, Yoram, & Willis, Robert J. 1997. Match quality, new information, and marital dissolution. *Journal of Labor Economics*, **15**(1, Part 2), S293–S329. [2], [5]
- Yoshida, Akiko. 2010. Role of Cultural Lag in Marriage Decline for Japan’s Boom and Bust Cohorts. *Marriage & Family Review*, **46**(1–2), 60–78. [9]

Zhang, Yuanting, & Van Hook, Jennifer. 2009. Marital dissolution among interracial couples. *Journal of Marriage and Family*, **71**(1), 95–107. [\[5\]](#)

A Appendix

A.1 Data Construction

This subsection outlines the construction of selected variables used in our analysis; variables with self-evident definitions are omitted for brevity.

Social Networks The JPSC asks for the total number of female and male friends of the respondent:

How many friends do you keep company with now? Answer this question based on how you met these individuals. The term “friend” here means a person with whom you keep personal company (or whom you meet, exchange letters with and talk with by wire), excluding a person with whom you keep company as the members of individual groups. If you had two or more first chances of company, choose the most important. These questions are asked in rounds 1 - 23. We impute these variables for round 24 onward by using their latest information available in round 23. We interpret these variables as proxies for respondents’ social networks.

Income Control Variables The JPSC collects detailed information on household income management, which allows us to identify the flow of money in the household. Through a series of questions, we know whether and what proportion of the income earned by each spouse goes to the husband’s purse, the wife’s purse, or the common purse, where the owner of a “purse” is the person who manages the income going there. To simplify the analysis, we define the shares of income controlled by the wife and jointly by the couple as the proportions of total household income that ultimately remain in the wife’s purse and the common purse, respectively.

We regard the amount of money going into the purse as the money managed by the owner. However, this amount does not represent the share of the money that the owner can use freely. For example, some of the money that went into the common purse could go to the wife’s or husband’s purse as pocket money. Let us use Figure A2 to visualize the flow of money in the household and explain how the income control variables are constructed. In the figure, the husband and wife earn 350 thousand yen and 150 thousand yen per month, respectively, for a total of 500 thousand yen per month. The husband puts 250 thousand yen and 100 thousand yen in the common purse and his own purse, respectively, whereas the wife puts her entire income in the common purse. The figure also shows that 30 thousand yen and 20 thousand yen flow out of the common purse to the husband and wife, respectively, as pocket money. Hence, the amount of money remaining in the common purse would be $250 + 150 - 30 - 20 = 350$, and the share of income controlled jointly is $350/500 = 0.7$ in our definition. Likewise, the share of income controlled by the wife is $20/500 = 0.04$.

B Additional Tables and Figures

B.1 Additional Tables

Table A1: Logistic Regression Predicting Non-Truncated Marriage Status

	Non-truncated (1)
Wife's Age at Marriage	1.166 (2.954)
Wife's Age at Marriage Squared	-0.020*** (0.003)
Husband's Age at Marriage	-0.165 (0.516)
Husband's Age at Marriage Squared	0.001 (0.002)
Husband older	0.383 (0.234)
Wife's Education at Marriage	1.004 (2.952)
Husband's Education at Marriage	-0.357 (0.502)
Ratio of Spouse's Education	-4.534** (1.936)
Husband's Birth Year	-0.071 (0.149)
Wife's Birth Year	0.197 (0.150)
Husband's Experience at Marriage	0.021 (0.497)
Wife's Experience at Marriage	0.581 (2.949)
Constant	-273.284*** (28.174)
Observations	2189
Pseudo R-squared	0.455
Log likelihood	-539.930

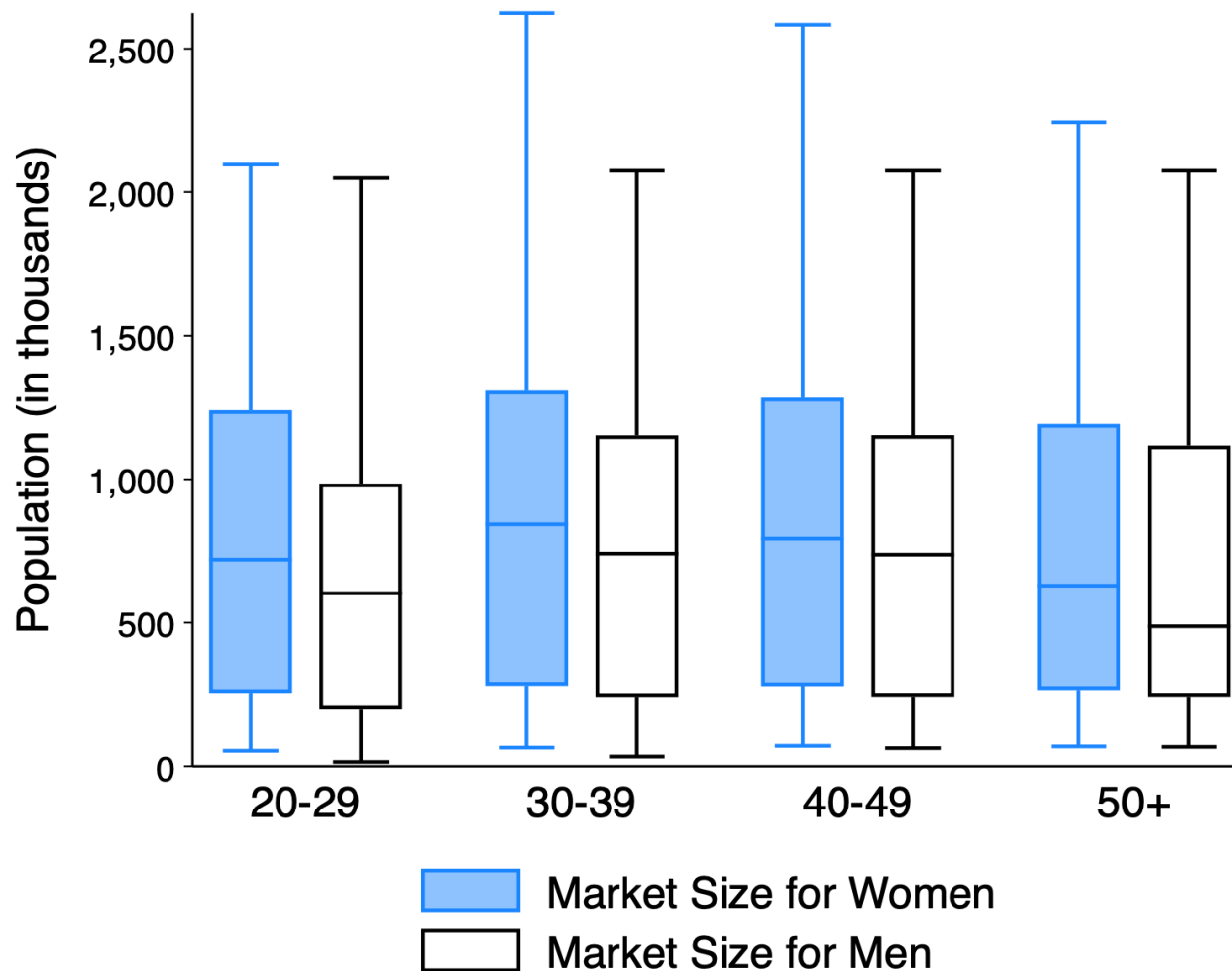
Note: The dependent variable equals one for non-truncated marriages and zero for truncated marriages. Standard errors are reported in parentheses. Region fixed effects are included. The model is used to construct the inverse probability weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Predicted Probabilities and Inverse Probability Weights

	Truncated			Non-truncated		
	N (1)	Mean (2)	SD (3)	N (4)	Mean (5)	SD (6)
Predicted Pr(non-truncated)	1821	0.094	0.175	368	0.536	0.249
Inverse probability weight	1821	1.256	1.095	368	3.202	4.521

Note: Predicted probabilities are obtained from a logistic regression of non-truncated status on observed marriage characteristics at the time of marriage. Inverse probability weights equal the inverse of the predicted probability of each marriage's observed truncation status.

B.2 Additional Figures

**Figure A1:** Marriage Market Size by Age and Gender

Note: Distribution of marriage market sizes across different age intervals for men and women. The boxes represent the interquartile range with the median marked inside each box, while the whiskers extend to the minimum and maximum values.

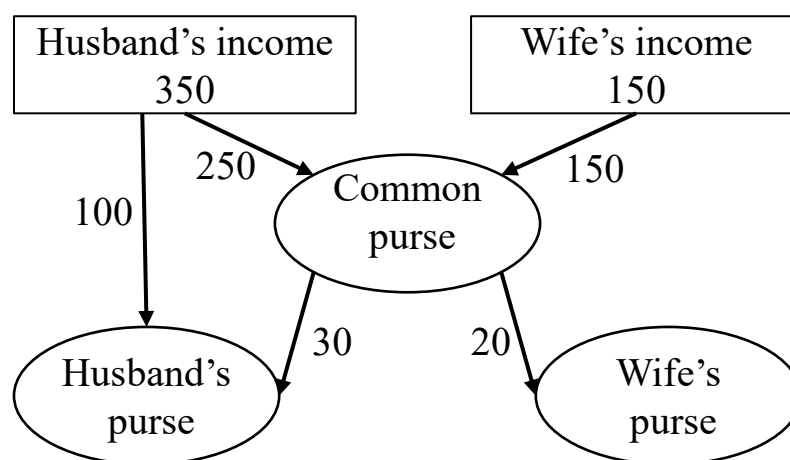


Figure A2: An example of the flow of money in the household. The numbers are in thousands of yen and are hypothetical.

OOB error across DynForest tuning parameters

Open marker shows best grid point: mtry = 12, nodesize = 3, minsplit = 2, OOB error = 4.926%

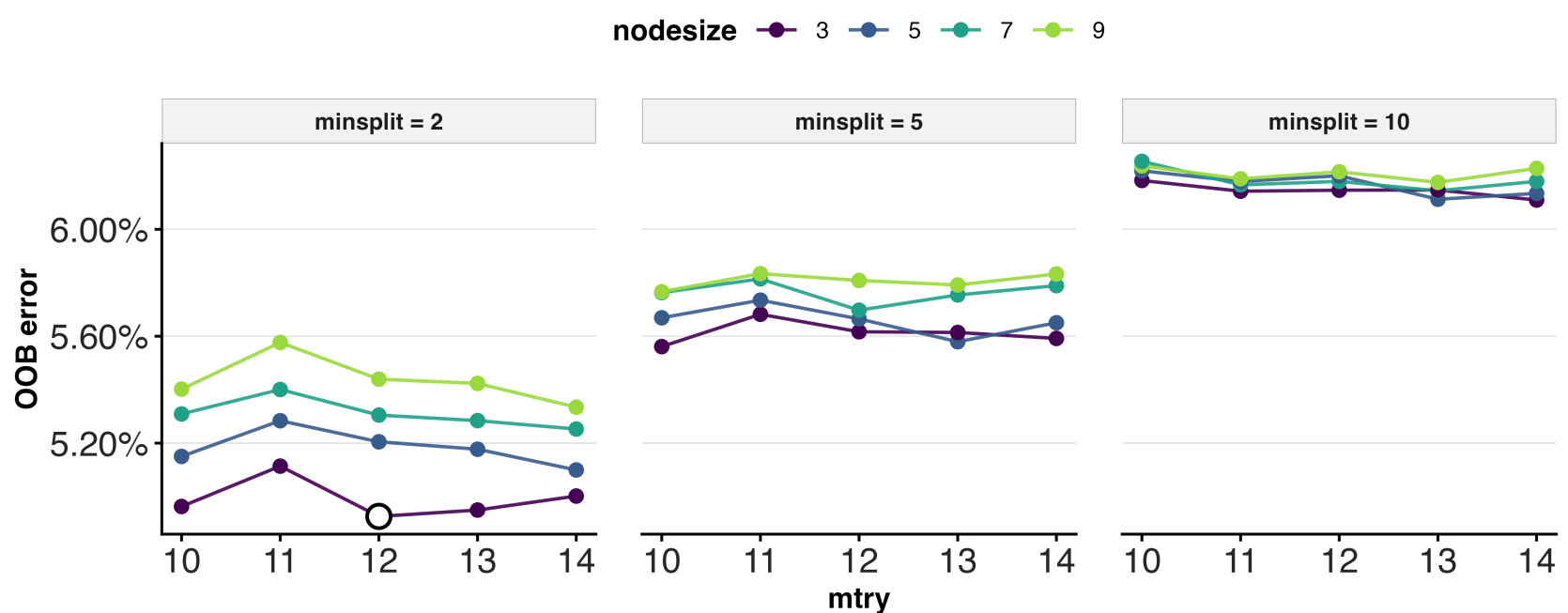


Figure A3: OOB Error Across DynForest Tuning Parameters

Note: The figure shows the out-of-bag prediction error from the DynForest tuning grid across values of the number of candidate variables, node size, and minimum split. Lines represent different node-size values, and panels represent different minimum-split values. The open circle marks the tuning combination with the lowest observed OOB error: number of candidate variables = 12, node size = 3, minimum split = 2, and OOB error = 4.926%.

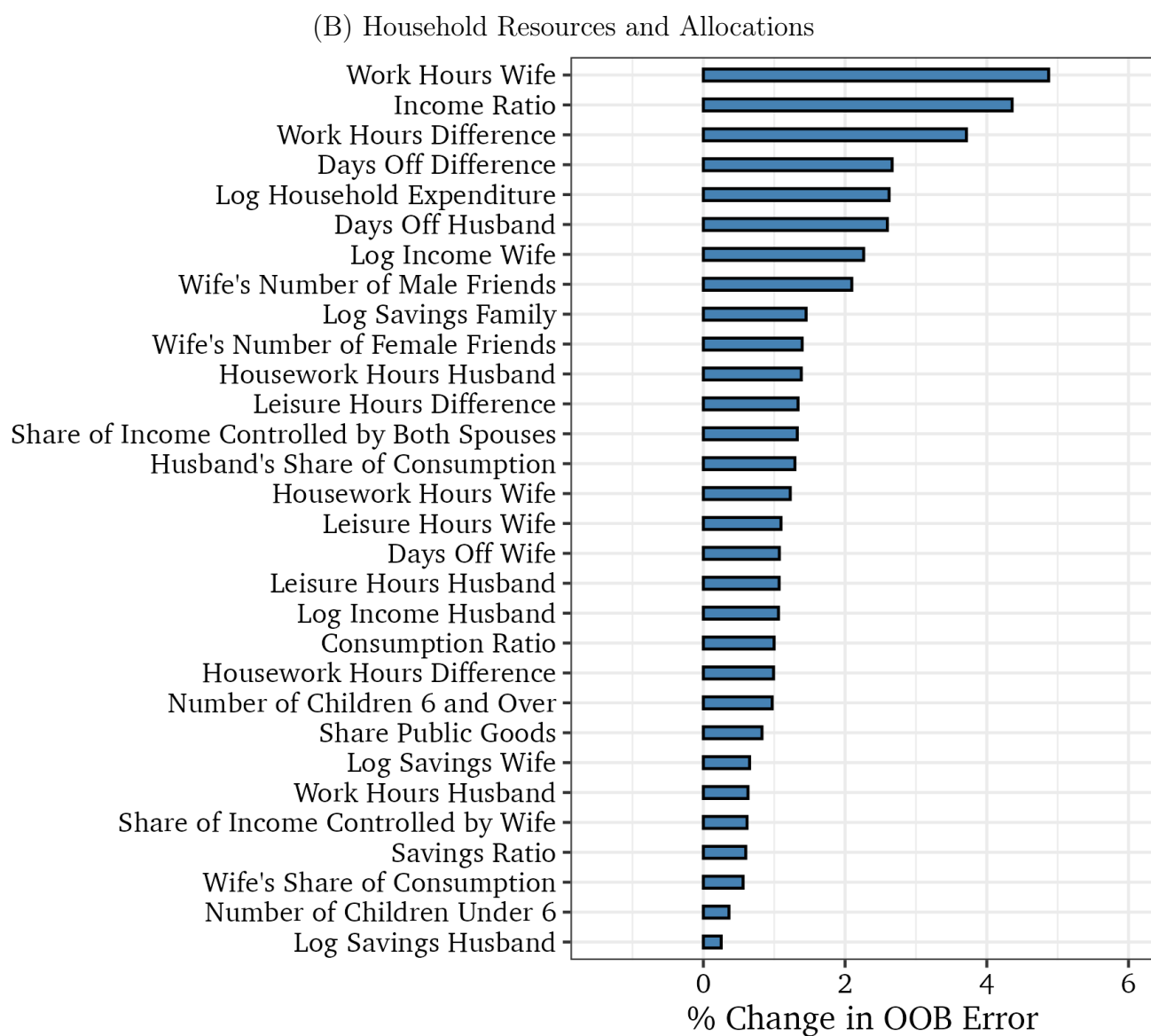
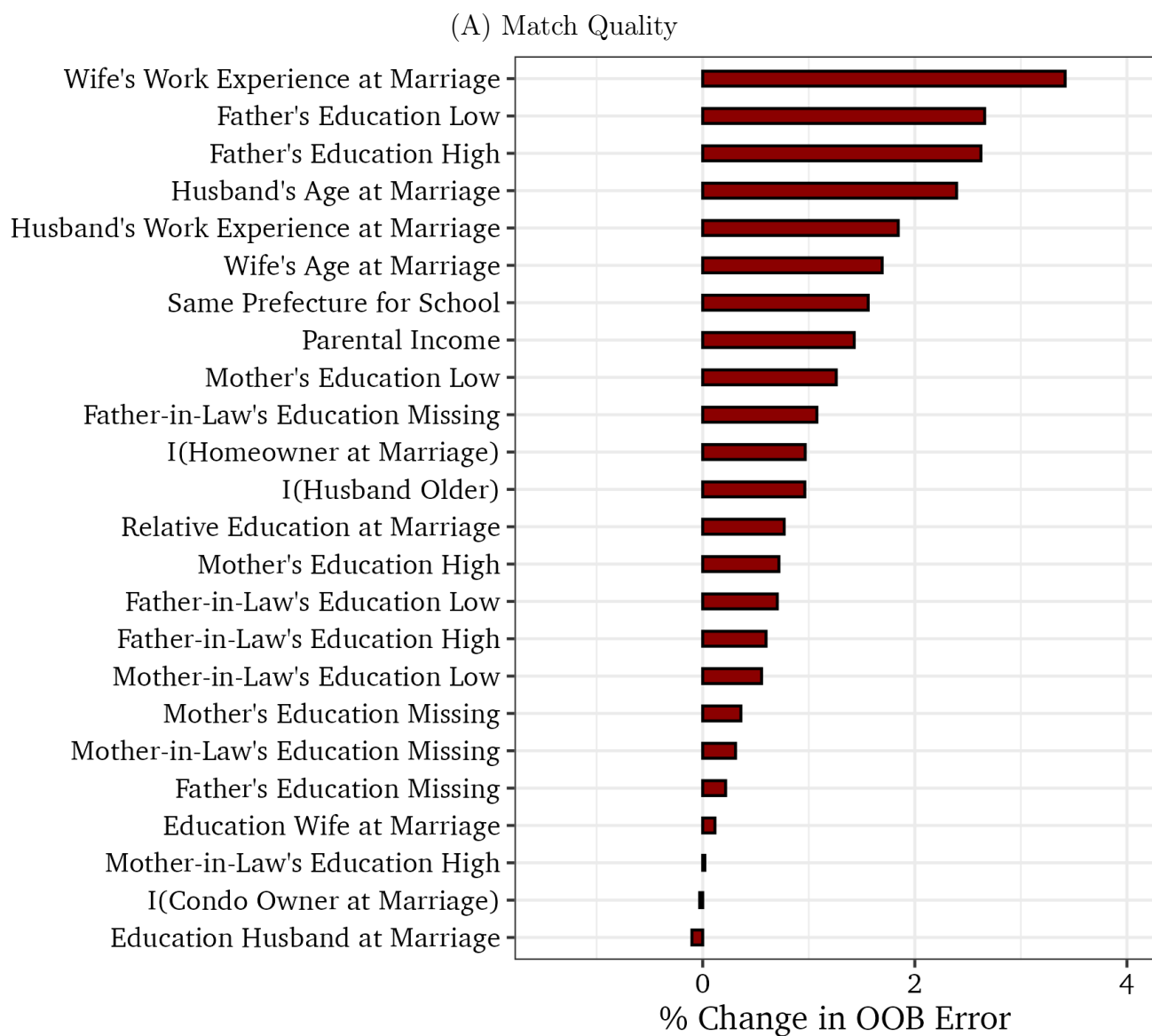


Figure A4: Variable Importance by Category

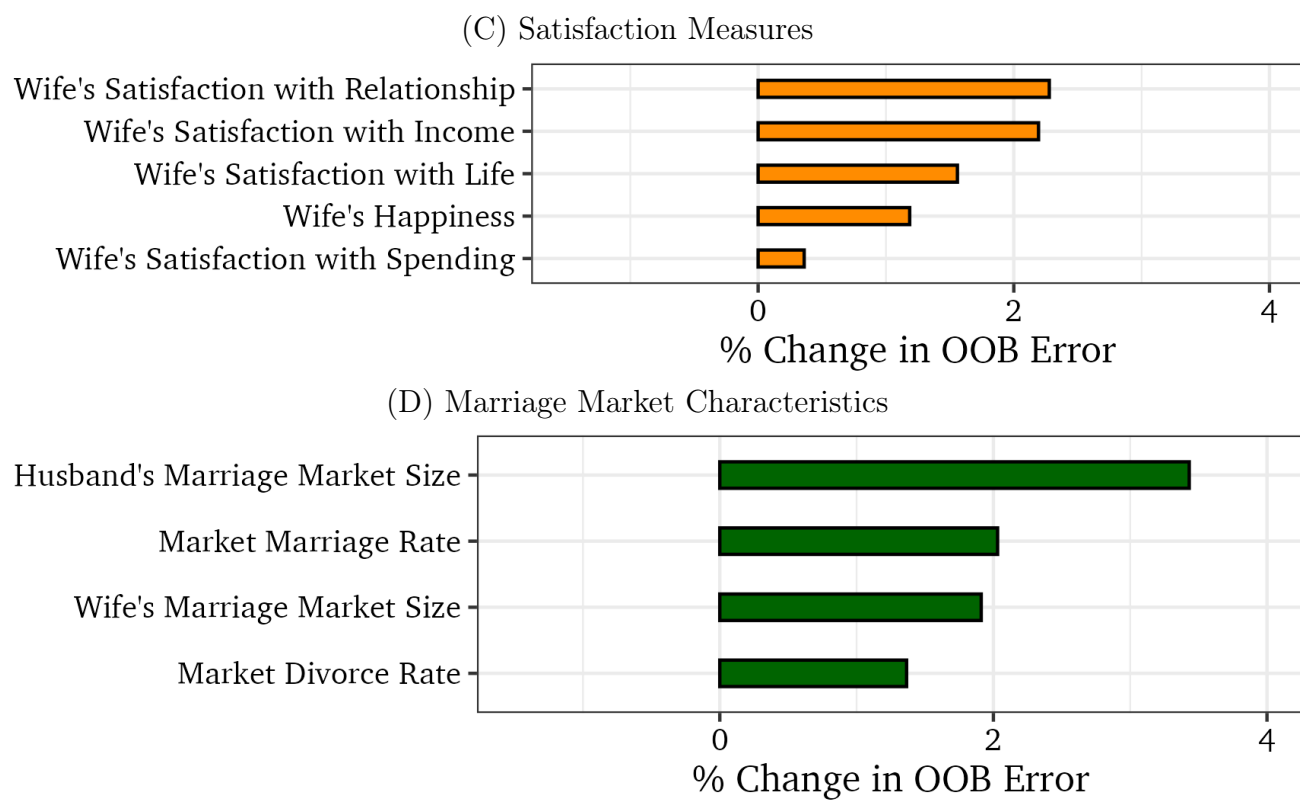


Figure A4: Variable Importance by Category

Note: Variable importance (VIMP) is displayed as the percent change in the out-of-bag error. We organize the sub-figures by variable category. Panel (A) includes match-quality variables measured at or near the time of marriage, Panel (B) includes household resources and allocations, Panel (C) presents satisfaction measures, and Panel (D) includes marriage-market characteristics. Positive values indicate that permuting the variable increases prediction error, and therefore that the variable contributes positively to predictive performance. Negative values indicate that permuting the variable lowers prediction error. We omit region variables in the interest of clarity. Estimates are obtained from an IPW-resampled DynForest model, where couples are sampled with replacement in proportion to their inverse-probability weights and each selected couple's full longitudinal history is retained.

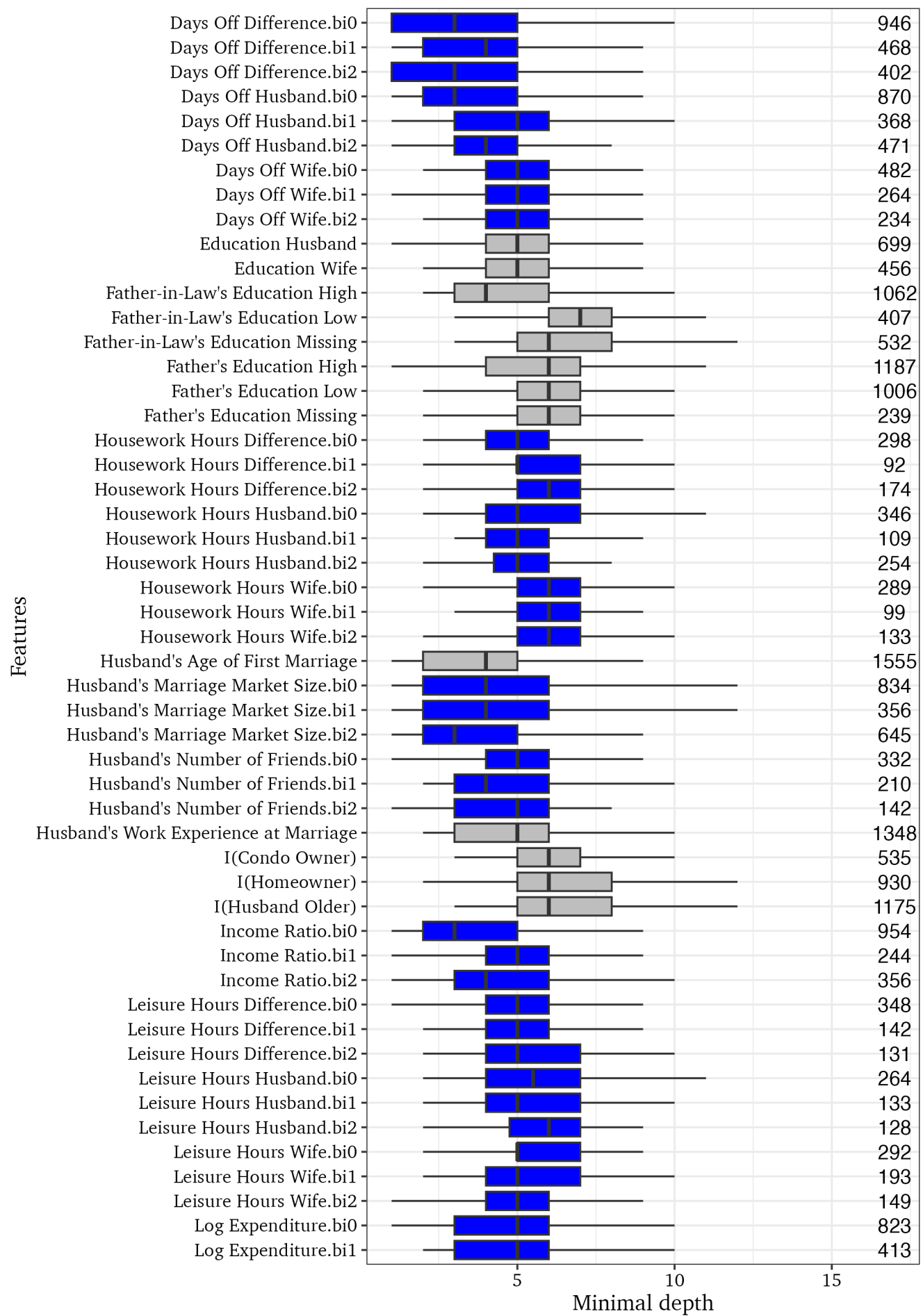


Figure A5: Minimal Depth

Note: Minimal depth computed when all variables are included across 2000 trees. Blue box plots denote time-varying features, while black box plots denote time-fixed features. Variables ending in “.b0”, “.b1”, and “.b2” denote the random intercept, linear slope, and quadratic terms, respectively. The number on the right-hand side indicates the number of trees (out of 2000) in which the variable appears. Variables with a lower minimum depth, and those that appear in more trees are considered more predictive.

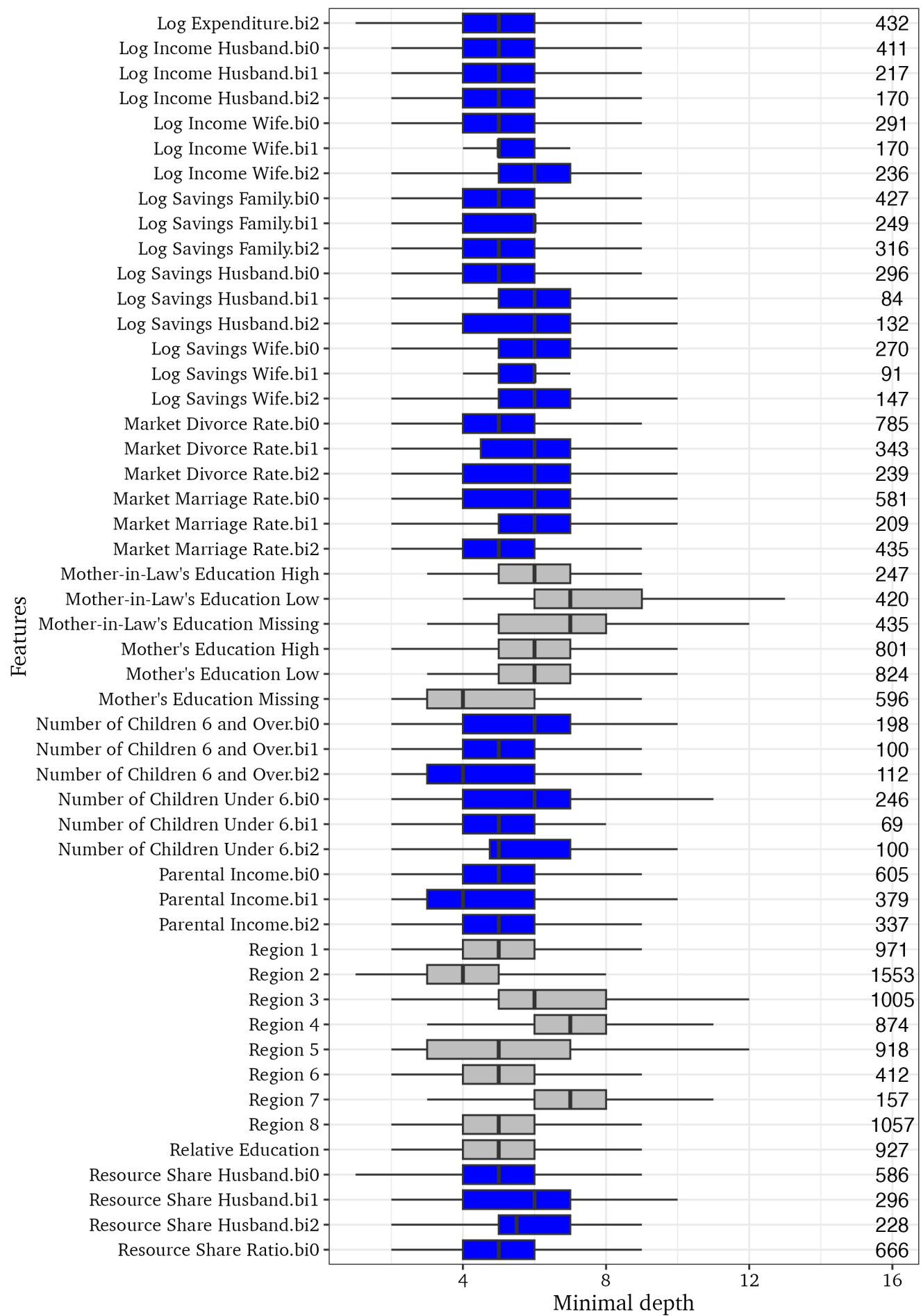


Figure A5: Minimal Depth

Note: Minimal depth computed when all variables are included across 2000 trees. Blue box plots denote time-varying features, while black box plots denote time-fixed features. Variables ending in “.b0”, “.b1”, and “.b2” denote the random intercept, linear slope, and quadratic terms, respectively. The number on the right-hand side indicates the number of trees out of 2000 that the variable is found. Variables with a lower minimum depth, and those that appear in more trees are considered more predictive.

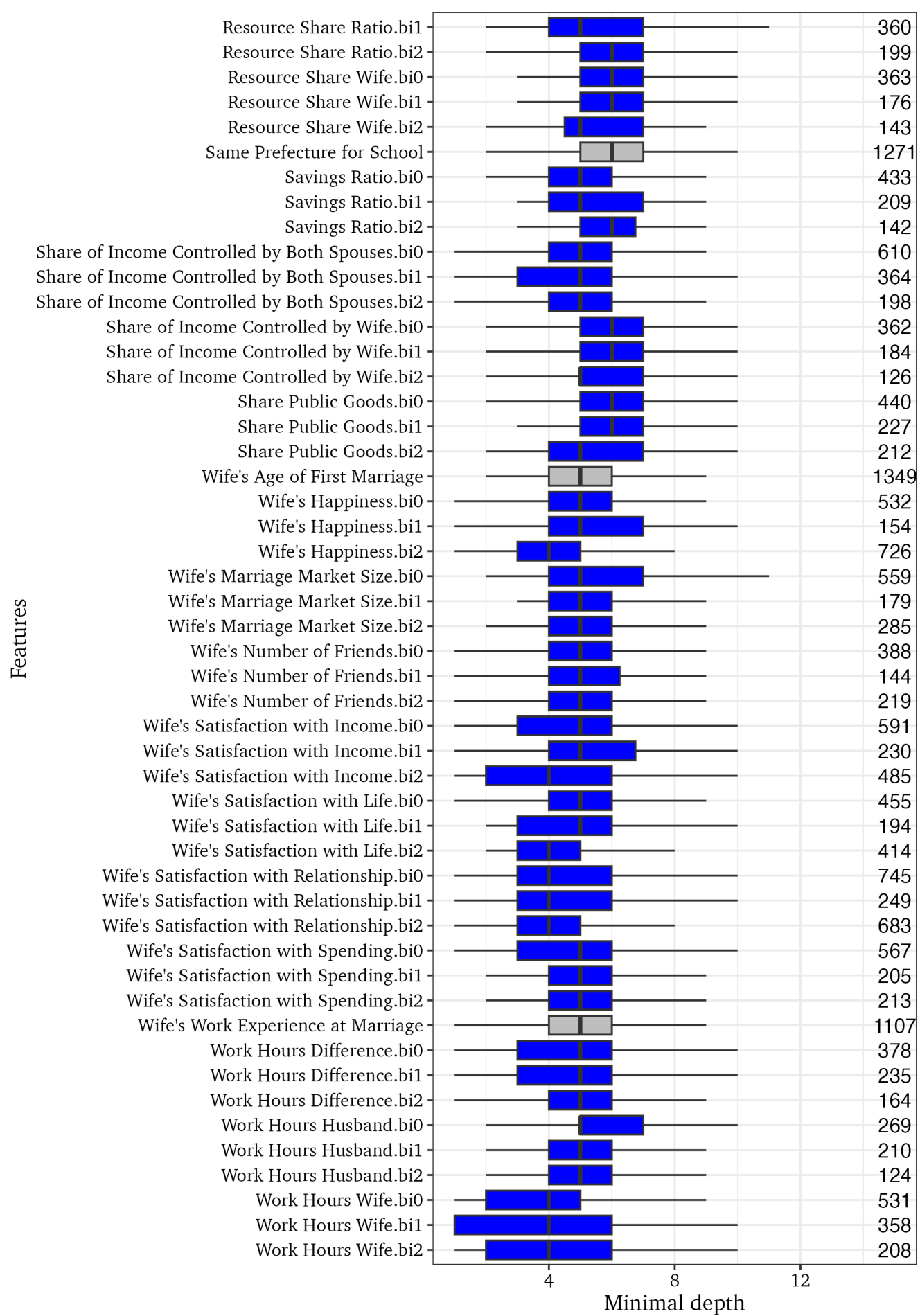


Figure A5: Minimal Depth

Note: Minimal depth computed when all variables are included across 2000 trees. Blue box plots denote time-varying features, while black box plots denote time-fixed features. Variables ending in “.b0”, “.b1”, and “.b2” denote the random intercept, linear slope, and quadratic terms, respectively. The number on the right-hand side indicates the number of trees out of 2000 that the variable is found. Variables with a lower minimum depth, and those that appear in more trees are considered more predictive.

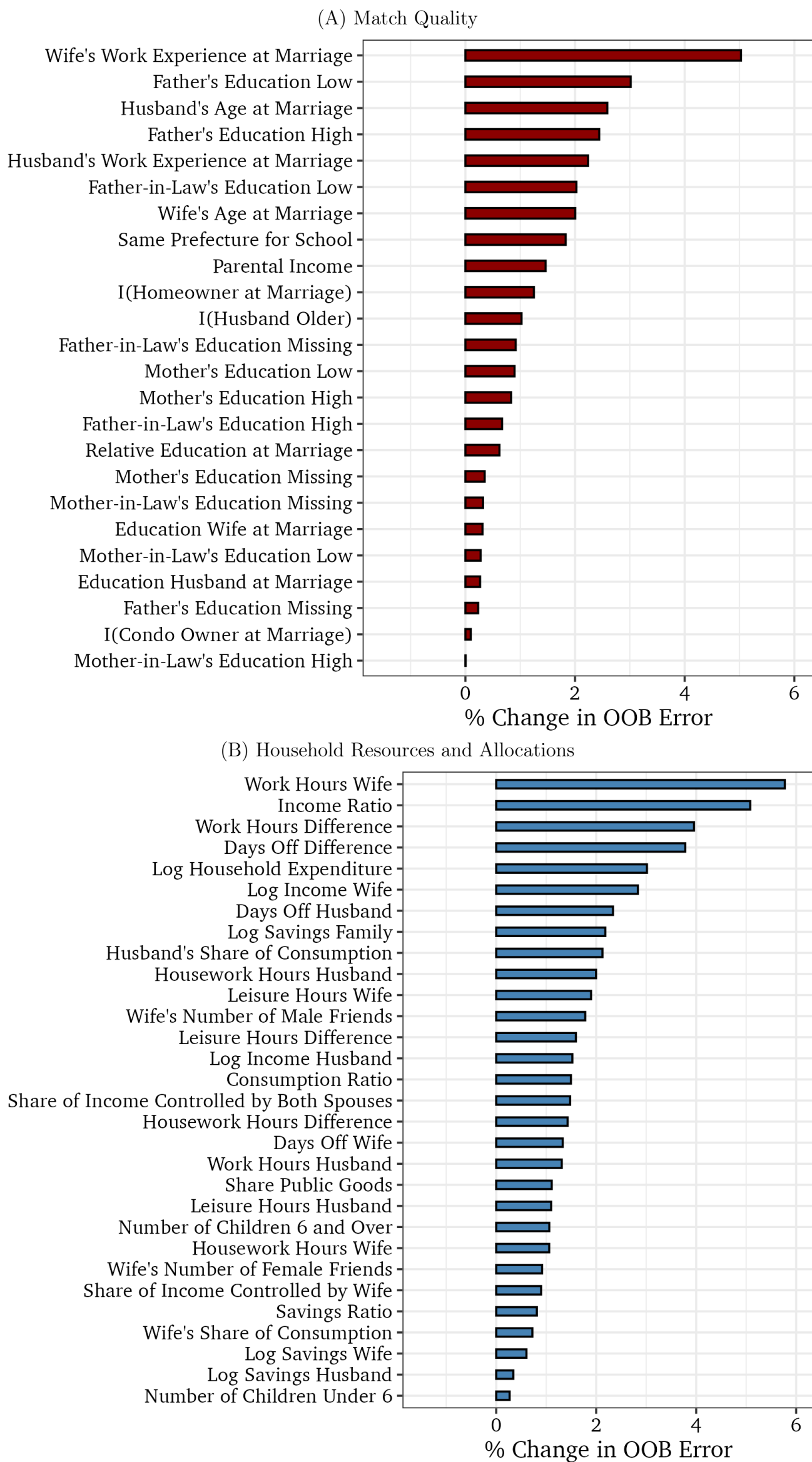


Figure A6: Variable Importance by Category (Omitting Satisfaction Measures)

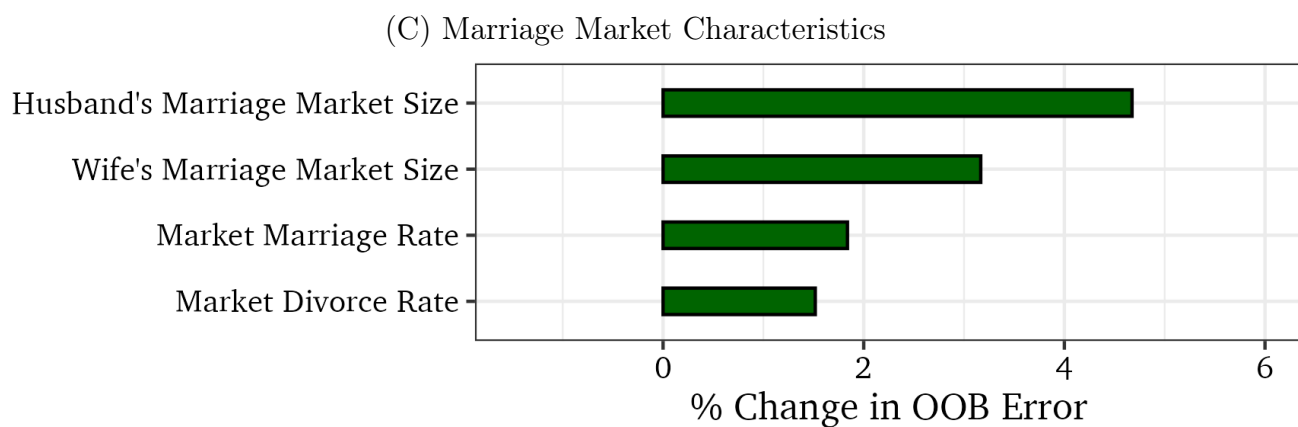


Figure A6: Variable Importance by Category (Omitting Satisfaction Measures)

Note: Variable importance (VIMP) is displayed as the percent change in the out-of-bag error. We organize the sub-figures by variable category. Panel (A) includes match-quality variables measured at or near the time of marriage, Panel (B) includes household resources and allocations, and Panel (C) includes marriage-market characteristics. Satisfaction measures are omitted from this specification. Positive values indicate that permuting the variable increases prediction error, and therefore that the variable contributes positively to predictive performance. Negative values indicate that permuting the variable lowers prediction error. We omit region variables in the interest of clarity. Estimates are obtained from an IPW-resampled DynForest model, where couples are sampled with replacement in proportion to their inverse-probability weights and each selected couple's full longitudinal history is retained.

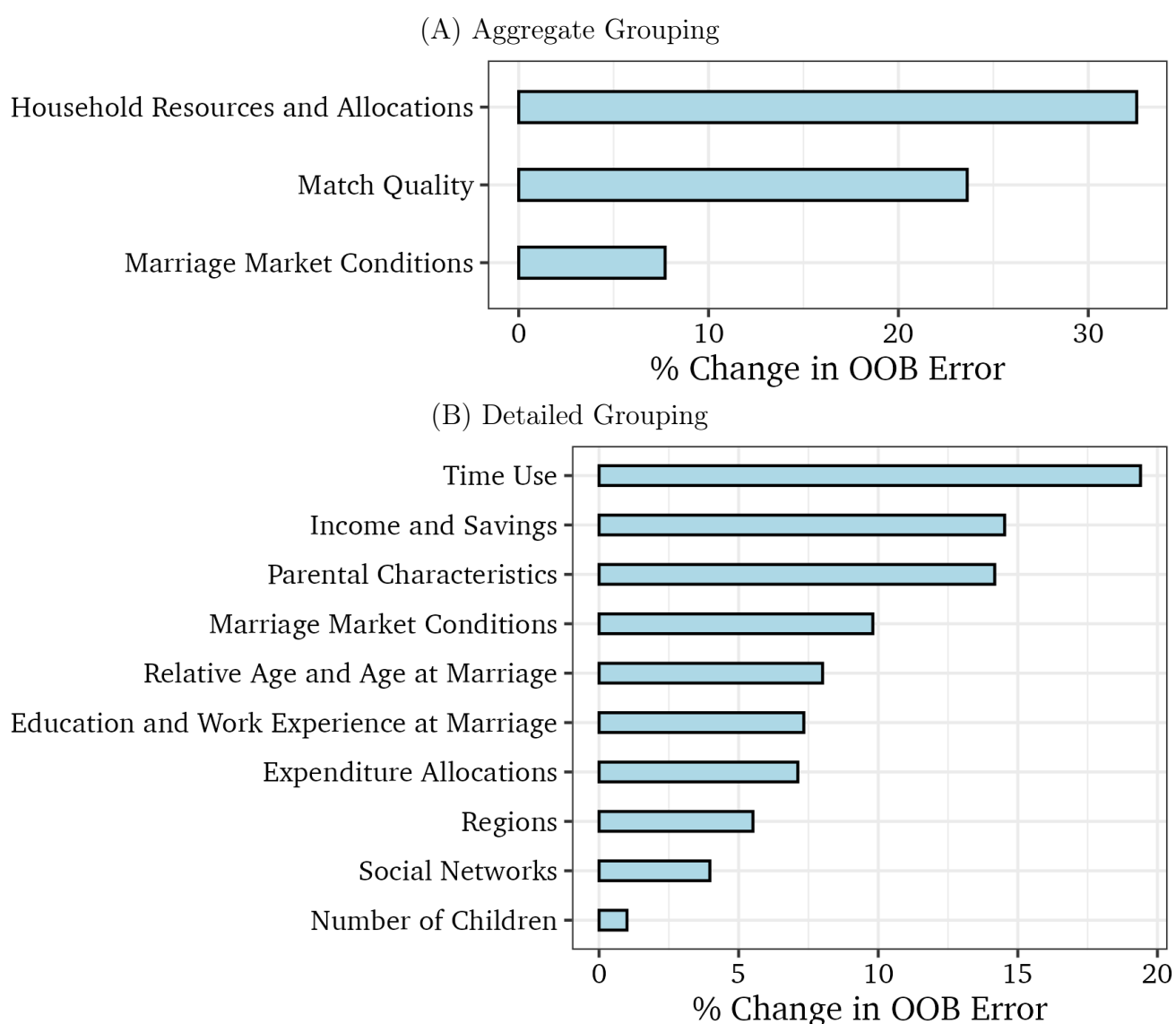


Figure A7: Group Variable Importance (Omitting Satisfaction Measures)

Note: Group Variable Importance (gVIMP). gVIMP is displayed as the percentage loss in the Out-of-Bag Error. In Panel A, we group variables following the sub-figures of Figure A4. In Panel B we disaggregate match quality and household resources and allocations into smaller categories of related variables. Income and savings include log individual incomes of both spouses, log individual and family savings, and relative income and savings. Expenditure allocations include all consumption variables, and the share of income managed by the wife and the couple jointly, respectively. Time use includes hours spent in market work, household work, and leisure, as well as differences in time spent in the categories across spouses. Time use also includes days off per week for both spouses. Social networks consist of the number of male and female friends for the wife. The remaining categories are self-explanatory. Estimates are obtained from an IPW-resampled DynForest model, where couples are sampled with replacement in proportion to their inverse-probability weights and each selected couple's full longitudinal history is retained.

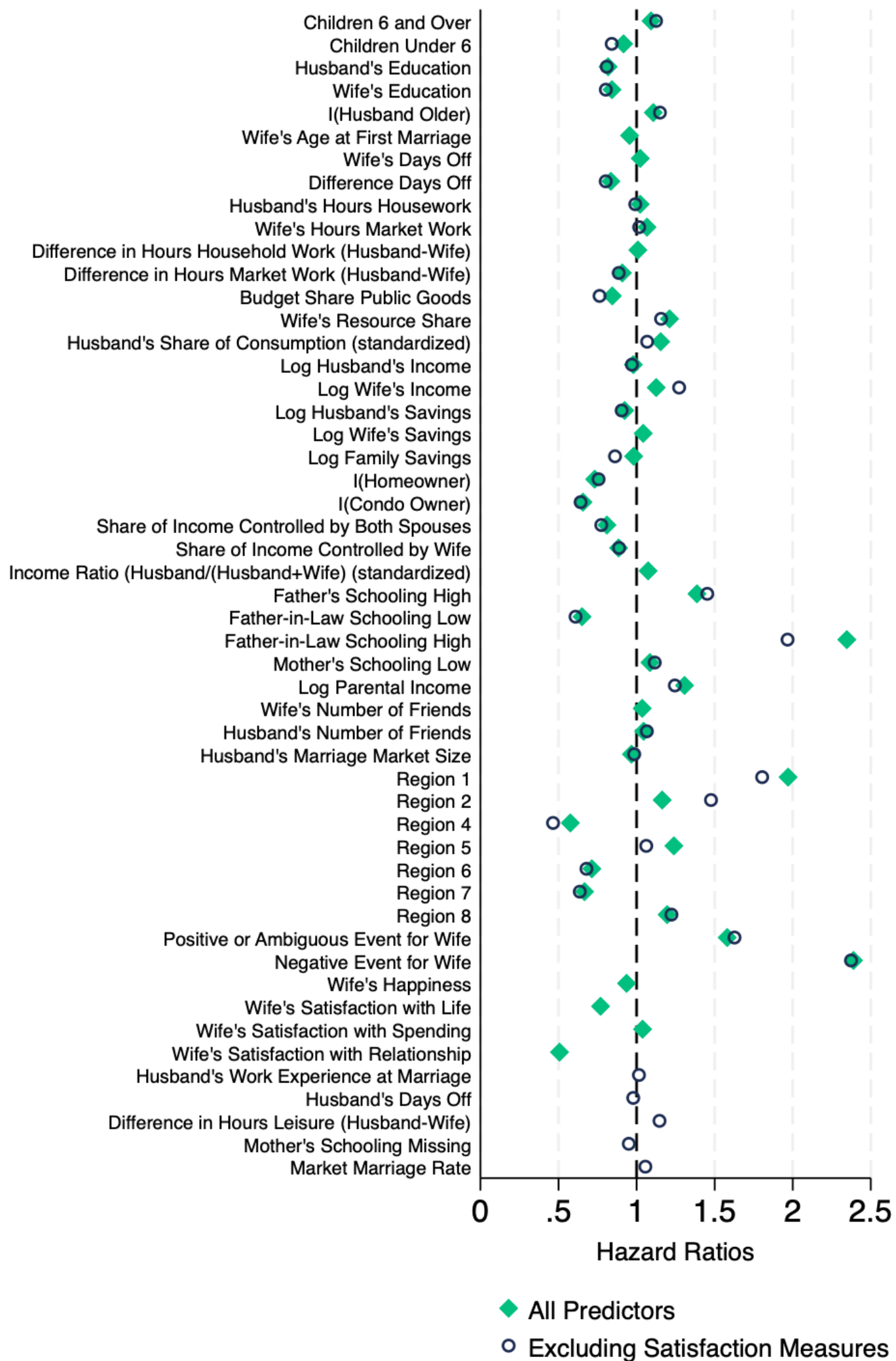


Figure A8: Cox Lasso Hazard Ratios with Inverse Probability Weights

Note: Hazard ratios from the Lasso Cox model. Green diamonds indicate the model included all possible predictors. Black circles indicate the model excluded satisfaction measures. Continuous variables are standardized. Hazard ratios below one indicate a decrease in risk of divorce. We omit region variables for conciseness. The data are weighted according to the probability that a marriage is observed from its start.

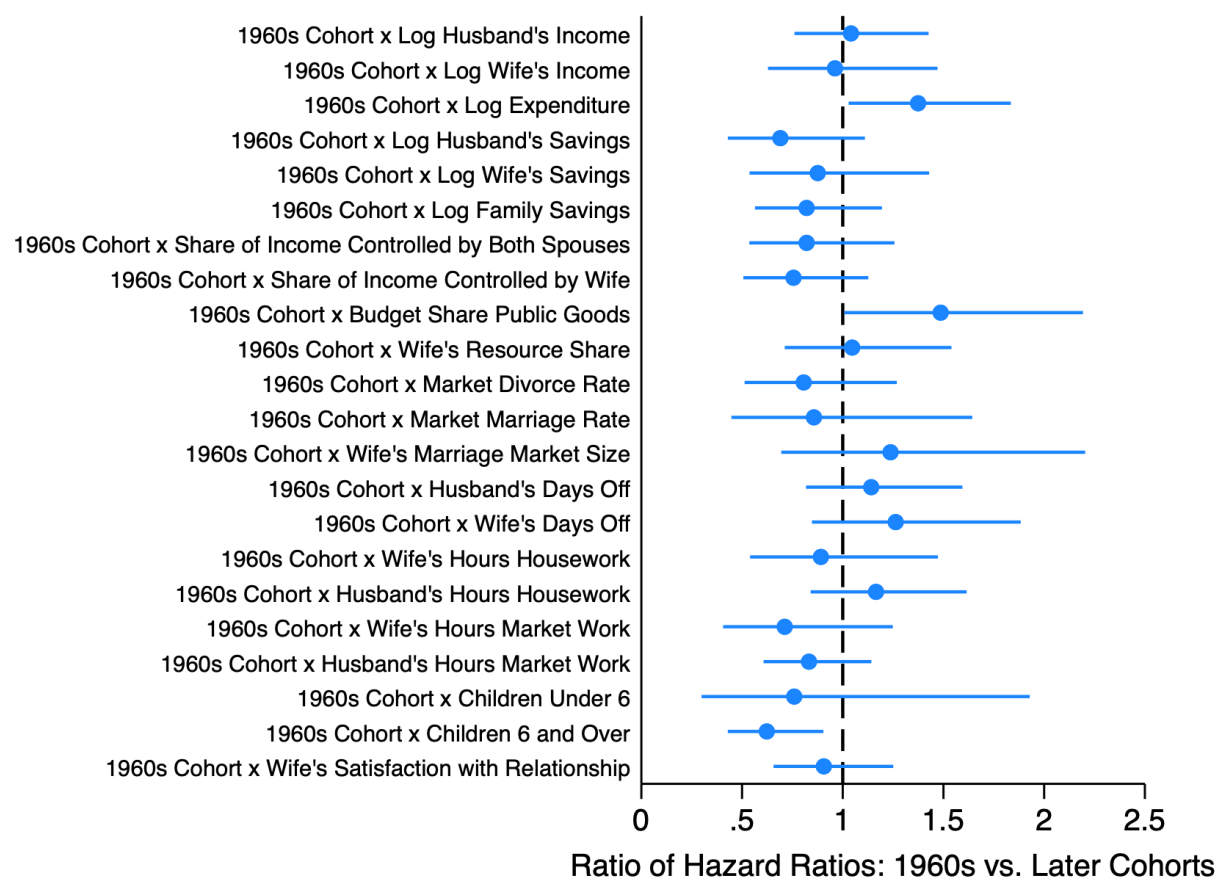


Figure A9: Cox Hazard Ratios by Cohort

Note: The figure reports interaction estimates from a Cox proportional hazard model in which selected predictors are interacted with an indicator for the wife belonging to the 1960s birth cohort. Estimates are plotted as ratios of hazard ratios for the 1960s cohort relative to later cohorts. Values above (below) one indicate that the association between the predictor and divorce hazard is larger (smaller) for the 1960s cohort than for later cohorts. Continuous variables are standardized to ease comparison across coefficients. Horizontal lines represent 95 percent confidence intervals.



Figure A10: Marriage Market Conditions and Divorce

Note: The figure reports hazard ratios from a Lasso Cox model that interacts marriage-market characteristics with the wife's reported satisfaction measures. Prediction 4 implies that favorable marriage-market conditions should be more strongly associated with divorce when match quality is low; therefore, because higher satisfaction indicates higher match quality, the interaction between marriage-market conditions and satisfaction should generally have a hazard ratio below one. Hazard ratios below one indicate a lower divorce hazard, while hazard ratios above one indicate a higher divorce hazard. The interaction estimates are mostly close to one and do not display a clear systematic pattern, providing limited support for Prediction 4.

C A Model of Dynamic Bargaining and Divorce

This appendix provides the mathematical details for the framework summarized in Section 2.2. The main text describes the economic motivation and states the empirical predictions. Here we formalize the dynamic bargaining problem, define the continuation region and divorce hazard, and derive the main comparative statics used in the predictions.

C.1 Preliminaries

Suppose that time is discrete and define the state vector $S_t = (K_t, \varepsilon_t, Y_t, M_t)$ at time t consisting of the marriage-specific capital K_t , match quality ε_t , spouses' incomes $Y_t = (Y_{ht}, Y_{wt})$, and marriage-market conditions $M_t = (M_{ht}, M_{wt})$, where Y_{it} and M_{it} for $i \in \{h, w\}$ are i 's income and local marriage-market conditions. A higher value of M_{it} represents a better prospect of remarriage. The match quality ε_t varies over time and follows a stationary AR(1) process:

$$\varepsilon_{t+1} = \rho\varepsilon_t + \eta_{t+1}, \quad \rho \in (0, 1), \quad (\text{A1})$$

where η_{t+1} is a mean-zero random variable that is independently and identically distributed, which empirically corresponds to an unobservable idiosyncratic shock to the couple. A fall in ε_t lowers expected future match quality because $\rho > 0$.

At the start of period t , predetermined variables (K_t, Y_t, M_t) are known. Then, match quality ε_t is realized and observed by both spouses. Conditional on $(K_t, \varepsilon_t, Y_t, M_t)$, spouses bargain over $A_t \equiv (C_{ht}, C_{wt}, G_t)$ —where C_{it} is the private-good expenditure for spouse $i \in \{h, w\}$ and G_t is the public-good expenditure—and then decide whether to remain married as elaborated below. Divorce in period t , therefore, occurs *after* observing ε_t in this model. As will become clear, the divorce decision is deterministic conditional on the realized ε_t , but stochastic at the start of period t before ε_t is realized.

C.2 Bargaining

Married couples allocate total household income $Y_{ht} + Y_{wt}$ between C_{it} for $i \in \{h, w\}$ and G_t subject to the budget constraint:

$$C_{ht} + C_{wt} + G_t = Y_{ht} + Y_{wt}. \quad (\text{A2})$$

The public good G_t captures shared expenditures and investments that increase future marital capital, including child-related spending. We assume that marriage-specific capital evolves according to:

$$K_{t+1} = (1 - \delta)K_t + \phi(G_t), \quad (\text{A3})$$

where $\delta \in (0, 1)$ and ϕ is twice continuously differentiable, strictly increasing, and weakly concave.

We denote the intra-household resource allocation by A_t and spouse i 's per-period utility in marriage by U_{it}^M :

$$U_{it}^M(A_t; S_t) = u_i(C_{it}) + v_i(G_t, K_t) + \varepsilon_t, \quad (\text{A4})$$

where u_i is twice continuously differentiable, strictly increasing, and strictly concave, and v_i is twice continuously differentiable, increasing in both arguments, and jointly concave in (G_t, K_t) .

We assume that each spouse has an outside option value $V_{it}^O(Y_{it}, M_{it})$. We assume that $\frac{\partial V_{it}^O}{\partial Y_{it}} > 0$ and $\frac{\partial V_{it}^O}{\partial M_{it}} > 0$. Thus, higher income and more favorable marriage-market conditions raise the spouse's threat point.

For any feasible allocation A_t satisfying (A2), define the value to spouse i from remaining married as

$$\tilde{V}_i^M(A_t, S_t) = u_i(C_{it}) + v_i(G_t, K_t) + \varepsilon_t + \beta \mathbb{E}[\max\{V_i^M(S_{t+1}), V_{i,t+1}^O\} \mid S_t, A_t], \quad (\text{A5})$$

where V_i^M is the equilibrium value of marriage defined below.

The continuation region $\mathcal{C}$, the Nash bargaining rule $A_t^*(S_t)$, and the equilibrium value functions $V_i^M(S_t)$ for each $i \in \{h, w\}$ are determined jointly as part of a recursive equilibrium. In a recursive equilibrium, they satisfy the following:

(i) for each $S_t \in \mathcal{C}$, the allocation $A_t^*(S_t)$ solves the Nash bargaining problem

$$A_t^*(S_t) \in \arg \max_{A_t \in F_t} \left(\tilde{V}_h^M(A_t, S_t) - V_{ht}^O \right) \left(\tilde{V}_w^M(A_t, S_t) - V_{wt}^O \right); \quad (\text{A6})$$

(ii) for each $S_t \in \mathcal{C}$, the equilibrium value of marriage is given by

$$V_i^M(S_t) = \tilde{V}_i^M(A_t^*(S_t), S_t); \quad (\text{A7})$$

(iii) the continuation region is given by

$$\mathcal{C} \equiv \left\{ S_t : \exists A_t \text{ satisfying (A2) such that } \tilde{V}_i^M(A_t, S_t) > V_{it}^O \text{ for both } i \in \{h, w\} \right\}.$$

For each state $S_t \in \mathcal{C}$, let F_t denote the set of allocations that satisfy the budget constraint and both spouses' strict participation constraints. We assume that F_t is compact and convex for every $S_t \in \mathcal{C}$, and that the Nash bargaining problem in (A6) admits a unique interior solution whenever $S_t \in \mathcal{C}$.

Thus, equations (A5)–(A7) jointly characterize the recursive equilibrium of the model on the continuation region. At any equilibrium allocation under marriage, the marital surplus

$$W_{it} \equiv V_i^M(S_t) - V_{it}^O$$

is strictly positive for each spouse $i \in \{h, w\}$. Divorce arises when $S_t \notin \mathcal{C}$, i.e., when no feasible allocation can satisfy both spouses' participation constraints. Since the current marriage-market conditions enter only through the current outside option, holding future state transitions fixed, we have:

$$\frac{\partial W_{it}}{\partial M_{it}} = -\frac{\partial V_{it}^O}{\partial M_{it}} < 0 \quad (\text{A8})$$

Under the model above, we define the discrete-time hazard as:

$$h_t(K_t, Y_t, M_t) \equiv \Pr(S_t \notin \mathcal{C} \mid K_t, Y_t, M_t), \quad (\text{A9})$$

where the probability on the right-hand side represents the probability of ε_t taking a value such that no feasible allocation

satisfies both spouses' participation constraints and budget constraint in period t , conditional on remaining married at the start of t . Then, letting T be the (random) divorce date, the discrete-time hazard is:

$$h_t = \Pr(T = t \mid T \geq t, \mathcal{I}_t),$$

where $\mathcal{I}_t$ denotes information available at the start of period t . Because time is discrete, h_t is a conditional probability rather than an instantaneous rate, and the cumulative probability of divorce over a horizon is obtained by iterating the hazard:

$$\Pr(T \leq t + k \mid \mathcal{I}_t) = 1 - \prod_{s=t}^{t+k} (1 - h_s).$$

In the data, divorce is observed between survey rounds. Although the empirical analysis uses Cox proportional hazards models, the annual survey data identify marital dissolution only over the interval between t and $t+1$. We therefore interpret the theoretical hazard $h_t(K_t, Y_t, M_t)$ as the model-implied conditional probability of divorce over that interval,

While we do not have a direct empirical counterpart of ε_t , the observed satisfaction measures can be interpreted as noisy proxies for realized match quality ε_t , while marriage-market variables and income components shift outside options V_{it}^O . Therefore, the empirical hazard regression can be interpreted as a reduced-form representation of the threshold condition, equation (A9).

C.3 Comparative statics

We conduct some comparative statics of the model described above and obtain the propositions below.²¹

Proposition 1 (Marriage-market conditions and divorce). *Fix $(K_t, Y_t, M_{-i,t})$ for $i \in \{h, w\}$, where $M_{-i,t}$ denotes the marriage-market conditions of the other spouse. Then an increase in spouse i 's marriage-market conditions M_{it} weakly increases the divorce hazard.*

Proposition 2 (Effect of match quality on continuation value). *At an interior solution, the married value function is increasing in match quality:*

$$\frac{\partial V_i^M(S_t)}{\partial \varepsilon_t} = 1 + \beta \rho \mathbb{E} \left[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\} \frac{\partial V_i^M(S_{t+1})}{\partial \varepsilon_{t+1}} \middle| S_t \right] > 0.$$

Proposition 3 (Marginal value of shared investment at a feasible allocation). *For any feasible allocation $A_t = (C_{ht}, C_{wt}, G_t)$, the marginal effect of shared investment G_t on spouse i 's value from remaining married is strictly positive:*

$$\frac{\partial \tilde{V}_i^M(A_t, S_t)}{\partial G_t} = \underbrace{\frac{\partial v_i(G_t, K_t)}{\partial G_t}}_{\text{current payoff}} + \underbrace{\beta \mathbb{E} \left[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\} \frac{\partial V_i^M(S_{t+1})}{\partial K_{t+1}} \middle| S_t, A_t \right]}_{\text{continuation shadow value}} \phi'(G_t) > 0. \quad (\text{A10})$$

C.4 From propositions to empirical predictions

The propositions above provide the formal basis for the predictions stated in Section 2.2. This subsection offers the mapping between the model and those predictions.

²¹The proofs of these propositions are provided in the Appendix D.

Prediction 1: Outside options and divorce risk. Proposition 1 shows that an improvement in spouse i 's marriage-market conditions M_{it} weakly increases the divorce hazard, since M_{it} enters the model through the outside option V_{it}^O . That is, a higher M_{it} raises the value of separation and tightens spouse i 's participation constraint. The continuation region therefore weakly shrinks, so the probability that the realized state falls outside the continuation weakly rises, which corresponds to Prediction 1 in the main text.

Prediction 2: Match quality and marital stability. Proposition 2 shows that the value of marriage is increasing in match quality ε_t . Therefore, adverse shocks to match quality reduce the value of marriage and increase the divorce hazard. In the empirical analysis, ε_t is not observed directly, and thus measures of relationship satisfaction are interpreted as noisy proxies for match quality. This gives Prediction 2.

Prediction 3: Shared investment and divorce hazard. Proposition 3 shows that public-good expenditure G_t has a positive marginal effect on the value of remaining married through two channels: current utility from the public good and the continuation value generated by higher future marriage-specific capital K_{t+1} . Thus, lower public-good expenditure reduces this investment channel and is associated with a lower future value of marriage, moving the couple closer to the boundary of the continuation region. This yields Prediction 3.

Prediction 4: Outside options matter more when match quality is low. Proposition 1 implies that better marriage-market conditions shrink the continuation region by raising outside options, while Proposition 2 implies that lower match quality reduces marital value. Intuitively, an improvement in marriage-market conditions raises a spouse's outside option in all states, but it is more likely to trigger divorce when the value of marriage is already low. In our model, this corresponds to a situation where ε_t is low, such that the couple is closer to the boundary at which no feasible allocation satisfies both participation constraints. Therefore, a given increase in M_{it} is more likely to move the realized state outside the continuation region when the match quality is lower. Using relationship satisfaction as a proxy for ε_t , we obtain Prediction 4.

D Proofs and technical details for reviewers

D.1 Existence and uniqueness of the dynamic value function

In this subsection, we prove the existence and uniqueness of the continuation value function conditional on a well-defined within-period bargaining rule $A^*(S)$. This is sufficient for the recursive formulation used in Appendix C, where the continuation region, bargaining rule, and value functions are jointly characterized as part of a recursive equilibrium.

Fix a within-period bargaining rule

$$A^*(S) = (C_h^*(S), C_w^*(S), G^*(S)),$$

assumed to be well-defined on the continuation region under the compactness, convexity, and interiority assumptions in Section C. For spouse i , define the induced current payoff under this policy as

$$\pi_i(S) \equiv u_i(C_i^*(S)) + v_i(G^*(S), K) + \varepsilon.$$

Assume $\pi_i(S)$ is bounded on $\mathcal{S}$.

Define the Bellman operator $T_i : \mathcal{B}(\mathcal{S}) \rightarrow \mathcal{B}(\mathcal{S})$ by

$$(T_i V)(S) = \pi_i(S) + \beta \mathbb{E}[\max\{V(S'), V_i^O(S')\} \mid S, A^*(S)], \quad (\text{A11})$$

where S' is next period's state induced by (A1), (A3).

For any $V_1, V_2 \in \mathcal{B}(\mathcal{S})$ and any S ,

$$|(T_i V_1)(S) - (T_i V_2)(S)| \leq \beta \mathbb{E}[|\max\{V_1(S'), V_i^O(S')\} - \max\{V_2(S'), V_i^O(S')\}| \mid S, A^*(S)].$$

Since $a \mapsto \max\{a, c\}$ is 1-Lipschitz for any constant c ,

$$|\max\{V_1(S'), V_i^O(S')\} - \max\{V_2(S'), V_i^O(S')\}| \leq |V_1(S') - V_2(S')| \leq \|V_1 - V_2\|_\infty.$$

Thus,

$$\|T_i V_1 - T_i V_2\|_\infty \leq \beta \|V_1 - V_2\|_\infty.$$

Because $\beta \in (0, 1)$, T_i is a contraction. By the Banach fixed-point theorem, there exists a unique bounded fixed point $V_i^M \in \mathcal{B}(\mathcal{S})$ conditional on the bargaining rule $A^*(S)$. This yields the unique continuation value function associated with that policy rule.

D.2 Proof of Proposition 1

Let $\mathcal{C}(M_t)$ denote the continuation region at marriage-market conditions $M_t = (M_{ht}, M_{wt})$, i.e.

$$\mathcal{C}(M_t) \equiv \left\{ S_t : \exists A_t \text{ satisfying (A2) such that } \tilde{V}_i^M(A_t, S_t) > V_{it}^O \text{ for both } i \in \{h, w\} \right\}.$$

Fix (K_t, Y_t, M_{ht}) and consider two values $M'_{wt} > M_{wt}$. Since $\partial V_{wt}^O / \partial M_{wt} > 0$, we have

$$V_{wt}^O(Y_{wt}, M'_{wt}) > V_{wt}^O(Y_{wt}, M_{wt}).$$

Therefore, for any feasible allocation A_t , if the wife's participation constraint is satisfied under M'_{wt} ,

$$\tilde{V}_w^M(A_t, S_t) > V_{wt}^O(Y_{wt}, M'_{wt}),$$

then it is also satisfied under M_{wt} . The converse need not hold. Hence the set of allocations satisfying both spouses' participation constraints weakly shrinks as M_{wt} rises, which implies

$$\mathcal{C}(M_{ht}, M'_{wt}) \subseteq \mathcal{C}(M_{ht}, M_{wt}).$$

By the hazard definition in (A9),

$$h_t(K_t, Y_t, M_{ht}, M_{wt}) = \Pr(S_t \notin \mathcal{C}(M_t) \mid K_t, Y_t, M_t).$$

Since the continuation region weakly shrinks when M_{wt} rises, the event $S_t \notin \mathcal{C}(M_t)$ becomes weakly more likely. Thus,

$$\frac{\partial h_t(K_t, Y_t, M_t)}{\partial M_{wt}} \geq 0.$$

The same argument yields

$$\frac{\partial h_t(K_t, Y_t, M_t)}{\partial M_{ht}} \geq 0.$$

□

D.3 Proof of Proposition 2

At the equilibrium allocation $A_t^*(S_t)$, equations (A5) and (A7) imply

$$V_i^M(S_t) = u_i(C_{it}^*) + v_i(G_t^*, K_t) + \varepsilon_t + \beta \mathbb{E}[\max\{V_i^M(S_{t+1}), V_{i,t+1}^O\} \mid S_t, A_t^*(S_t)].$$

Differentiating with respect to ε_t and using $\varepsilon_{t+1} = \rho\varepsilon_t + \eta_{t+1}$, so that $\partial\varepsilon_{t+1}/\partial\varepsilon_t = \rho$, yields

$$\frac{\partial V_i^M(S_t)}{\partial \varepsilon_t} = 1 + \beta\rho \mathbb{E}\left[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\} \frac{\partial V_i^M(S_{t+1})}{\partial \varepsilon_{t+1}} \mid S_t\right].$$

To establish positivity, define

$$D(S_t) \equiv \frac{\partial V_i^M(S_t)}{\partial \varepsilon_t}.$$

Then D satisfies the functional equation

$$D(S_t) = 1 + \beta\rho \mathbb{E}[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\} D(S_{t+1}) \mid S_t].$$

Consider the operator

$$(\mathcal{T}D)(S_t) \equiv 1 + \beta\rho \mathbb{E}[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\}D(S_{t+1}) \mid S_t] .$$

Since $\beta\rho \in (0, 1)$ and the indicator function is bounded by one, $\mathcal{T}$ is a contraction on the space of bounded functions under the sup norm. Therefore, $\mathcal{T}$ admits a unique bounded fixed point D , where

$$D(S_t) = \frac{\partial V_i^M(S_t)}{\partial \varepsilon_t}.$$

To establish positivity, note that if $f(S) \geq 0$ for all S , then

$$(\mathcal{T}f)(S_t) = 1 + \beta\rho \mathbb{E}[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\}f(S_{t+1}) \mid S_t] \geq 1.$$

Thus $\mathcal{T}$ maps nonnegative bounded functions into strictly positive bounded functions. Since the fixed point D is unique, it follows that

$$D(S_t) \geq 1 > 0 \quad \text{for all } S_t.$$

Therefore,

$$\frac{\partial V_i^M(S_t)}{\partial \varepsilon_t} > 0.$$

This proves Proposition 2. □

D.4 Proof of Proposition 3

Fix a feasible allocation $A_t = (C_{ht}, C_{wt}, G_t)$ and state S_t . By definition,

$$\tilde{V}_i^M(A_t, S_t) = u_i(C_{it}) + v_i(G_t, K_t) + \varepsilon_t + \beta \mathbb{E}[\max\{V_i^M(S_{t+1}), V_{i,t+1}^O\} \mid S_t, A_t] . \quad (\text{A12})$$

Differentiate (A12) with respect to G_t . Since $u_i(C_{it})$ and ε_t do not depend on G_t , we obtain

$$\frac{\partial \tilde{V}_i^M(A_t, S_t)}{\partial G_t} = \frac{\partial v_i(G_t, K_t)}{\partial G_t} + \beta \frac{\partial}{\partial G_t} \mathbb{E}[\max\{V_i^M(S_{t+1}), V_{i,t+1}^O\} \mid S_t, A_t] . \quad (\text{A13})$$

The choice G_t affects the continuation term only through next period's marriage-specific capital:

$$K_{t+1} = (1 - \delta)K_t + \phi(G_t),$$

so

$$\frac{\partial K_{t+1}}{\partial G_t} = \phi'(G_t).$$

Applying the chain rule to the continuation term in (A13) gives

$$\begin{aligned}
\frac{\partial}{\partial G_t} \mathbb{E}[\max\{V_i^M(S_{t+1}), V_{i,t+1}^O\} \mid S_t, A_t] &= \mathbb{E}\left[\frac{\partial}{\partial G_t} \max\{V_i^M(S_{t+1}), V_{i,t+1}^O\} \mid S_t, A_t\right] \\
&= \mathbb{E}\left[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\} \frac{\partial V_i^M(S_{t+1})}{\partial K_{t+1}} \frac{\partial K_{t+1}}{\partial G_t} \mid S_t, A_t\right] \\
&= \mathbb{E}\left[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\} \frac{\partial V_i^M(S_{t+1})}{\partial K_{t+1}} \mid S_t, A_t\right] \phi'(G_t).
\end{aligned} \tag{A14}$$

Substituting (A14) into (A13), we obtain

$$\frac{\partial \tilde{V}_i^M(A_t, S_t)}{\partial G_t} = \underbrace{\frac{\partial v_i(G_t, K_t)}{\partial G_t}}_{\text{current payoff}} + \underbrace{\beta \mathbb{E}\left[\mathbf{1}\{V_i^M(S_{t+1}) > V_{i,t+1}^O\} \frac{\partial V_i^M(S_{t+1})}{\partial K_{t+1}} \mid S_t, A_t\right]}_{\text{continuation shadow value}} \phi'(G_t), \tag{A15}$$

which is the expression in Proposition 3.

To show strict positivity, note first that

$$\frac{\partial v_i(G_t, K_t)}{\partial G_t} > 0$$

by assumption. In addition, $\phi'(G_t) > 0$, and the continuation term is weakly nonnegative because it is multiplied by an indicator function and $\partial V_i^M(S_{t+1})/\partial K_{t+1} \geq 0$: an increase in marriage-specific capital weakly raises future marital value because $v_i(G, K)$ is increasing in K and K_{t+1} enters the continuation value through future utility. Therefore,

$$\frac{\partial \tilde{V}_i^M(A_t, S_t)}{\partial G_t} > 0.$$

This proves Proposition 3. □